An entry–exit law for the resurrection of dead neurons under smoothed ReLU activation

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Abstract

A neuron dies when its pre-activation margin is negative on every datum, and under hard ReLU its gradient is exactly zero, so it never returns. In the sharp-activation limit of a shallow softplus network under gradient flow we prove the dynamics is instead hybrid with memory: the dead unit is released exactly when the signed time integral of the residual excluding it returns to zero, at rate $O(\log\beta/\beta)$ and with a first-order release coefficient in which the arrival speed has cancelled. A sweep of 600 instances isolates 57 that satisfy the hypotheses; the law holds in all 57, with median relative error 2.7% over coefficients spanning a factor of 20. A pre-registered ladder follows: a reduced stochastic-schedule model is falsified, two instruments are negatives, and a confirmatory MLP run of 400 runs reproduces resurrection, the area in fact turning positive in 13 of 16 returns, hard ReLU structurally zero, the return probability monotone in β. A transformer confirmatory of 800 pre-registered runs transfers the class to the feed-forward block through the unit's own channel ($\hat{p} = 0.440$) and reproduces a second, upstream channel in all four arms: hard-ReLU units return 20 of 100 against 0 of 100 with attention frozen.

Keywords: entry–exit function; geometric singular perturbation; dead neurons; smoothed ReLU; gradient flow; loss of plasticity.

MSC 2020: Primary 34E15; Secondary 34A36, 68T07.

1 Introduction

Write $\Phi : \mathcal{W} \to \mathcal{F}$ for the parametrization of a network and $J = D\Phi$ for its Jacobian. A neuron whose pre-activation margin is negative on every datum contributes an identically zero column to J: it sits on a stratum of the singular locus where J drops rank. This is exact for ReLU and exponentially thin for softplus-β; it does not hold for GELU at all, whose output is nonzero for every finite negative margin (Section 6.6). Under a hard ReLU the gradient of the loss in the directions of that unit is exactly zero, so the stratum is absorbing.

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A dead unit stays dead. This is the mechanism behind the empirically documented dormant-neuron and loss-of-plasticity phenomena [1, 2, 3]. The question of this paper is what replaces the absorbing stratum when the activation is smoothed, as it is in every softplus or GELU [4] network actually trained.

Our answer is an exact gating law and its numerical arc.

Theorem (informal)**.** In the sharp-activation limit of the smoothed flow, a dead unit does not sit on an absorbing stratum: it freezes and is released exactly when the signed time integral of the residual that excludes it returns to zero. The stratum is a memory gate rather than a trap. The convergence to that hybrid limit is uniform away from the two switching instants, at the rate $O(\log\beta/\beta)$ in the activation stiffness β, and the release time carries the first-order coefficient $-1/(cR(t_{\text{exit}}))$, in which the arrival speed has cancelled: the residual at death drops out, while the coefficient still depends on the datum constant c and on the residual at the exit. The precise statement is Theorem 2.

Scope of the proved object. The theorem covers the minimal case: one unit, one active datum at a time, a *fixed* readout and a one-dimensional input. The experiments of Section 6 train the readout v_k along with everything else, so the object that is proved and the object that is measured differ in that dimension. The transfer between them is empirical, not deduced.

The contributions are:

- **A theorem (Section 3 and Theorem 2).** In the sharp limit $\beta \to \infty$ of a shallow softplus network under gradient flow, the death and resurrection of a neuron is the entry–exit problem of a slow–fast system [5]: the dead unit freezes and reactivates when $\int R\,dt = 0$, where R is the residual excluding that unit. The uniform rate is $O(\log\beta/\beta)$ and the release time carries the first-order coefficient $-1/(cR(t_{\text{exit}}))$, from which the arrival speed has cancelled. The proof is organized as six lemmas (Section A), each with an independent executable symbolic or numerical verification. A consequence of the same construction is that the two limit orders, smooth-then-sharp against the pointwise generalized-gradient flow, do not commute (Section 4): the field-first order is the frozen branch of Eberle, Jentzen, Riekert and Weiss [6], our flow-first limit resurrects, and the autodiff convention that fixes the branch is discussed in Section 5. For the erf smoothing we verify numerically that the release criterion is unchanged; the proof is written for softplus, and GELU is outside the class because its derivative is non-monotone.
- **A sweep from one instance to a law (Section 3.4).** Every constant is verified on a reference two-neuron instance, and a sweep of 600 random instances then tests the release law against the dynamics on the 57 that satisfy the hypotheses: it holds in all 57, with a median relative error of 2.68% over predicted coefficients ranging from -91.8 to -4.5.

- **A pre-registered experimental ladder (Section 6).** Organized by prediction, with the null controls and the negatives shown: a reduced deterministic-schedule model falsified under SGLD (Section 6.1); two instrument designs recorded as negatives because a calibrated null fired (Section 6.2); the mechanism first observed in an MLP pressure pilot (Section 6.3); and a confirmatory MLP run of 400 runs in which the area law reproduces with confidence intervals, hard ReLU acts as the zero-gradient control, and the return probability is monotone in β (Section 6.4 and Figure 5). The scoping of the transformer (Section 6.6) exposed two new mechanisms, a GELU sign-flip barrier and an upstream channel, and its own pre-registered confirmatory of 800 runs was executed: the theorem class transfers to the feed-forward block through the own channel, the upstream channel reproduces in all four arms including hard ReLU, and the barrier remains confounded with injection depth.
- **A confirmatory in a transformer block (Sections 6.5 and 6.6 and Section C).** The 800 pre-registered runs close their five estimands: the theorem class transfers to the feed-forward block through the unit's own channel (22/50 sustained returns, the area opening before 21/22 of them); the upstream channel reproduces in all four arms, hard ReLU included (20/100 against 0/100 with attention frozen); and the GELU barrier is left confounded with injection depth under the pre-registered three-way label. Table 4 carries the five rows.

The theorem is proved for the minimal configuration above and its constants are verified on one reference instance and on a sweep of 600 drawn instances; the two confirmatories, the MLP (4a.3) and the transformer (`etapa-5`[1]), cover two small architectures at zero noise; the stochastic regime is open and, where we probed it, it breaks the deterministic schedule (Section 6.1).

The paper is organized as follows. Section 2 fixes the model and the four hypotheses. Section 3 states the entry–exit theorem, the six lemmas that compose it and the numerical verification of its constants; the complete proofs of the lemmas are deferred to Section A. Section 4 makes the non-commutation of the two limit orders precise and connects it to the plasticity literature, which Section 5 surveys. Section 6 is the experimental ladder, from the falsified reduced model to the two executed confirmatories, the MLP and the transformer; Sections 7 to 9 give the limitations, the conclusion, and the reproducibility record.

2 Setup and notation

Data $(x_i, y_i)_{i=1}^n \in \mathbb{R} \times \mathbb{R}$ in general position; shallow network with unit output weights

$$f_\theta(x) = \sum_k g_\beta(w_k x + b_k), \qquad g_\beta(z) = \beta^{-1} \log(1 + e^{\beta z}), \tag{1}$$

[1]Artifact, commit and script names are in Portuguese, the project's lab language; they are kept verbatim because the commit hashes and manifests reference them.

quadratic loss $L = \frac{1}{2n}\sum_i (f_\theta(x_i) - y_i)^2$ and gradient flow $\dot\theta = -\nabla L$ on the training horizon $[0, H]$. General monotone sigmoids are treated in Section 3. The parameter β is the *activation stiffness*: as $\beta \to \infty$, $g_\beta \to \mathrm{ReLU}$ and $g'_\beta \to \mathbf{1}_{\{z>0\}}$. We write H for the training horizon throughout, a time in the flow and a number of optimizer steps in the experiments, and $\mathcal{N}$ for the standard normal cumulative distribution function.

Fix one neuron A and one datum j. Its *margin* is $z = \theta_A \cdot (x_j, 1)$; write $c = \|(x_j, 1)\|^2/n$ and let

$$R(t) = y_j - f_{-A}\big(x_j; \theta_B(t)\big) \tag{2}$$

be the *residual excluding* A, where θ_B collects the other parameters. The *hybrid reference* θ^{hyb} runs the full ReLU flow until the first time t_{hit} with $z = 0$, then freezes θ_A and runs θ_B alone; set R^{hyb} accordingly and

$$\mathcal{A}^{\mathrm{hyb}}(t) = \int_{t_{\mathrm{hit}}}^{t} R^{\mathrm{hyb}}(s)\, ds \tag{3}$$

for its signed area. The death of A is exactly the rank drop of J on the stratum $\{z \le 0\}$; the question is the fate of the trajectory on that stratum.

Hypotheses. On the episode we assume

$R^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$ (transversal death); (H1)

$\mathcal{A}^{\mathrm{hyb}}$ has a first zero at $t_{\mathrm{exit}} < H$ with $R^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$ and $\mathcal{A}^{\mathrm{hyb}} < 0$ on $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$; (H2)

the other margins of A stay $\le -\delta_1 < 0$; (H3)

no other switching event occurs in $[0, H]$. (H4)

3 The entry–exit theorem

Definition 1 (hybrid limit)**.** θ^∞ equals: the ReLU flow on $[0, t_{\mathrm{hit}}]$; on $[t_{\mathrm{hit}}, t_{\mathrm{exit}}]$, $\theta_A \equiv \theta_A(t_{\mathrm{hit}})$ (frozen) with θ_B the B-only flow; on $[t_{\mathrm{exit}}, H]$, the ReLU flow restarted at $(\theta_A(t_{\mathrm{hit}}), \theta_B^{\mathrm{hyb}}(t_{\mathrm{exit}}))$.

Theorem 2 (entry–exit law for the smoothed dead unit)**.** *Assume* (H1)–(H4)*. Then for every* $\delta > 0$

$$\sup_{t \in [0,H] \setminus (B_\delta(t_{\mathrm{hit}}) \cup B_\delta(t_{\mathrm{exit}}))} \big\|\theta^\beta(t) - \theta^\infty(t)\big\| = O_\delta\!\left(\frac{\log\beta}{\beta}\right), \tag{4}$$

and the first return $t^\beta_{\mathrm{release}}$ *of the margin* z *to* 0 *after death satisfies*

$$t^\beta_{\mathrm{release}} = t_{\mathrm{exit}} - \frac{1}{c\, R^{\mathrm{hyb}}(t_{\mathrm{exit}})} \cdot \frac{1}{\beta} + o(\beta^{-1}). \tag{5}$$

Remark 3 (what carries each constant). The $\log\beta$ in (4) is sharp and is carried solely by the normal coordinate of θ_A inside the episode, whose sticking depth is $z \sim -\beta^{-1}\log\beta$ (Lemma 8); θ_B obeys $O(\log\beta/\beta^2)$ (Lemma 6) and the tangential components of θ_A are

exponentially frozen (Lemma 10). In the first-order coefficient of (5) the arrival speed cancels: in the product of the entry delay by the area it cuts, the residual at death drops out and only the reciprocal exit speed survives, so each corner contributes the reciprocal of its own transversal speed (Lemma 9). What does not cancel is the dependence on c and on $R^{\mathrm{hyb}}(t_{\mathrm{exit}})$, which are instance quantities. The remainder is $o(\beta^{-1})$; the numerics suggest $O(\log\beta/\beta^2)$, which would follow from quantifying the $o(1)$ of Lemma 7 with $\delta_z = \log\beta/\beta$, and we leave it as a stated refinement. Figure 1 draws the three stages this remark describes. The limit is that of the regularized flow; discrete ReLU gradient descent freezes the dead neuron forever (Section 4).

Remark 4 (which hypotheses are essential). (H1)–(H2) are essential: they are the entry–exit geometry itself, and the law fails without the sign change of R^{hyb}. (H3)–(H4) are there to simplify the proof: (H3) makes the other data contribute $O(e^{-\beta\delta_1})$ and (H4) isolates a single episode. We expect both to be removable at the cost of bookkeeping; we have not done it.

3.1 The six lemmas

The episode reduces to a scalar entry–exit problem through the six lemmas below, proved in Section A. Each also carries a committed executable symbolic or numerical verification (`passo0--6`, `prova_L1--L6`), whose outcome is reported at the end of each proof.

Lemma 5 (normal form and additive quadrature)**.** *Let $w = e^{\beta z}$ and $u = 1/w$, and set $\Psi(u) = \log u + u$. In fast time $\tau = \beta t$ the margin obeys*

$$\frac{dw}{d\tau} = c\,R(t)\,\frac{w^2}{1+w}\,(1+E_\beta), \qquad |E_\beta| \le \frac{\log(1+w)}{\beta|R|}, \tag{6}$$

the quadratic normal form of De Maesschalck and Schecter [5, equations (1.9)–(1.10)] with $\varepsilon = \beta^{-1}$, and one has the exact additive identity

$$\frac{d\Psi(u)}{dt} = -c\,\beta\,R(t) + c\log(1+w) + \mathcal{E}_{(\mathrm{H3})}(t), \qquad \left|\mathcal{E}_{(\mathrm{H3})}\right| \le C_3\,\beta\,e^{-\beta\delta_1}. \tag{7}$$

Under (H1)–(H2) the additive correction satisfies $\int_{t_{\mathrm{hit}}}^{t_{\mathrm{release}}} c\log(1+w)\,ds = O(\log\beta/\beta)$.

The bound $|E_\beta| \le \log(1+w)/(\beta|R|)$ in (6) degenerates at the interior zero of R that (H1)–(H2) force, since R changes sign inside the episode; that is precisely why the proof switches to the quadrature Ψ of (7), which never divides by R.

Lemma 6 (B-subsystem and feedback)**.** *Under (H4) the B-field is affine with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, where G_B is the Gram matrix of the active data, and R^{hyb} is affine in θ_B. Integral Grönwall consuming the bound of Lemma 5 gives, conditionally on the layer being maintained (supplied by Lemma 9),*

$$\sup\left\|\theta_B^{\beta} - \theta_B^{\mathrm{hyb}}\right\| = O\left(\frac{\log\beta}{\beta^2}\right), \tag{8}$$

one full order better than the naive $O(\beta^{-1})$: in the bulk $u = O(\beta|\mathcal{A}|)$, so the coupling is $O(\beta^{-2})$ there and $O(\beta^{-1})$ only in the $O(\beta^{-1})$-wide corner windows.

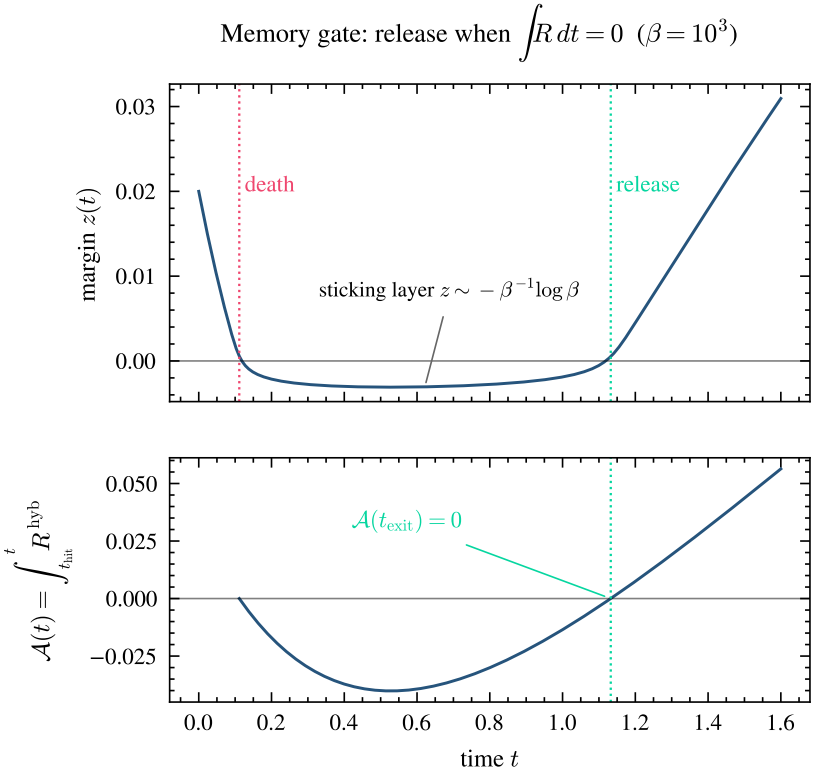


Figure 1: The memory gate at $\beta = 10^3$ (coupled system, reference instance). Top: the margin $z(t)$ dies at $t_{\rm hit}$, hugs the sticking layer of depth $\sim \beta^{-1}\log\beta$, and releases. Bottom: the hybrid area $\mathcal{A}(t) = \int_{t_{\rm hit}}^{t} R^{\rm hyb}$; release coincides with $\mathcal{A} = 0$ up to $O(\beta^{-1})$ (Theorem 2). Deterministic Euler integration ($dt = 10^{-5}$); the uncertainty is discretization only. Script `t2-prova/figuras_draft.py`.

Lemma 7 (entry corner)**.** *Let $v = c\,|R^{\rm hyb}(t_{\rm hit})|$ be the arrival speed, transversal by (H1). The exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$ collapses the crossing-time difference to an elementary integral and gives*

$$t^{\beta}_{\rm cross} = t_{\rm hit} + \frac{1}{\beta v}\big(1 + O(\delta_z) + o(1)\big), \qquad \big\|\theta^{\beta}(t_{\rm cross}) - \theta^{\infty}(t_{\rm hit})\big\| = O(\beta^{-1}), \tag{9}$$

where $t^{\beta}_{\rm cross}$ is the first time with $w = w_0$. The exponentially weighted approach integral is $\frac{3/2}{\beta v}(1 + o(1))$, the factor $3/2$ coming from $\int_0^{\infty} e^{-s}(1+e^{-s})\,ds$.

Lemma 8 (sticking and release for prescribed R)**.** *Let $R \in C^1$ be prescribed and satisfy (H1)–(H2), and write $\mathcal{A}(t) = \int_{t_{\rm hit}}^{t} R$. Then*

(i) $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + \mathrm{corr}(t)$ with $0 \le \mathrm{corr}(t) = \int c\log(1+w)\,ds = O(\log\beta/\beta)$; on compact subintervals of $(t_{\rm hit}, t_{\rm exit})$, $u = c\beta|\mathcal{A}(t)|\big(1 + O(\log\beta/(\beta a_0))\big)$, so the sticking depth is $z \sim -\beta^{-1}\log\beta$;

(ii) the release obeys the closed formula

$$t_{\text{release}} - t_{\text{exit}} = \frac{\text{corr}(\beta)}{c\,\beta\,R(t_{\text{exit}})}\,(1+o(1)) = O\!\left(\frac{\log\beta}{\beta^2}\right), \tag{10}$$

so u returns to $O(1)$ at $t_{\text{exit}} + O(\beta^{-1})$, with no $\log\beta$ factor;

(iii) a perturbation $\tilde{R} = R + \delta$ shifts t_{exit} by at most $\delta\,(t_{\text{exit}} - t_{\text{hit}})/R(t_{\text{exit}}) + O(\delta^2)$.

No division by R occurs anywhere: the layer is held by the area, not by the sign of R, and the interior zero of R forced by (H1)–(H2) is harmless.

Lemma 9 (coupled bootstrap)**.** *Under (H1)–(H4) the coupled system satisfies the two-term law*

$$t^{\beta}_{\text{release}} = t_{\text{exit}} - \frac{1}{c\,R^{\text{hyb}}(t_{\text{exit}})}\cdot\frac{1}{\beta} + \frac{\text{corr}(\beta)}{c\,\beta\,R^{\text{hyb}}(t_{\text{exit}})} + o(\beta^{-1}), \qquad \text{corr}(\beta) = O\!\left(\frac{\log\beta}{\beta}\right), \tag{11}$$

obtained by continuation with strict margins, with the constant $M_2 = e^{L_B H} C_f C_4/c$ independent of the continuation hypothesis for $\beta \ge \beta_0$. In the first-order coefficient the arrival speed cancels: in the product $K\,|R^{\text{hyb}}(t_{\text{hit}})|/R^{\text{hyb}}(t_{\text{exit}})$ of the entry delay by the area it cuts, the factor $|R^{\text{hyb}}(t_{\text{hit}})|$ cancels against $K = 1/(c|R^{\text{hyb}}(t_{\text{hit}})|)$ of Lemma 7, leaving a coefficient that still depends on c and on $R^{\text{hyb}}(t_{\text{exit}})$.

Lemma 10 (exit corner, branch selection and scalar reduction)**.** *Under (H1)–(H4), after release: (i) the β-trajectory crosses the exit corner in time $\frac{1}{\beta v'}(1+o(1))$ with $v' = c\,R^{\text{hyb}}(t_{\text{exit}}) > 0$, the mirror of Lemma 7; (ii) the ReLU flow restarted at $(\theta_A(t_{\text{hit}}), \theta_B^{\text{hyb}}(t_{\text{exit}}))$ has forward uniqueness on the activating branch, and that branch is selected by the limit itself, the frozen Filippov branch [7] being incompatible with release at $t_{\text{exit}} + O(\beta^{-1})$; (iii) classical Grönwall glues on $[t_{\text{release}}, H]$ at rate $O(\log\beta/\beta)$. Under (H3) the other data contribute $O(e^{-\beta\delta_1})$, the face-j field is rank one along $(x_j, 1)$, and the episode reduces to the scalar problem.*

3.2 Proof of the theorem

Proof of Theorem 2. We compose the six lemmas along the four stages of the episode.

By Lemma 10 the field on θ_A splits under (H3) into a rank-one part along $(x_j, 1)$ and terms of size $O(e^{-\beta\delta_1})$, so the tangential components of θ_A are frozen at an exponential rate and the normal component is exactly the scalar problem of Lemmas 5 to 9.

On $[0, t_{\text{hit}} - \delta]$ both flows are the ReLU flow up to the softplus defect, which is bounded by $Ce^{-\beta z^0(t)}$; step 1 of the proof of Lemma 7 integrates that defect against the trajectory and gives a gap $O(\beta^{-1})$. Lemma 7 then delivers the entry corner: the crossing time is $t_{\text{hit}} + 1/(\beta v)(1+o(1))$ and the state at crossing is $O(\beta^{-1})$ from $\theta^{\infty}(t_{\text{hit}})$.

On $[t_{\text{hit}} + \delta, t_{\text{exit}} - \delta]$, Lemma 9 closes the feedback loop: its continuation argument keeps the margin strictly inside the layer and θ_B within $O(\log\beta/\beta^2)$ of the hybrid, by consuming Lemma 6 (which needs the layer) and Lemma 8(iii) (which tolerates the resulting perturbation of the forcing). Inside the layer Lemma 8(i) gives the depth $z \sim -\beta^{-1}\log\beta$, and this is the largest of the three contributions to (4); θ_B contributes $O(\log\beta/\beta^2)$ and

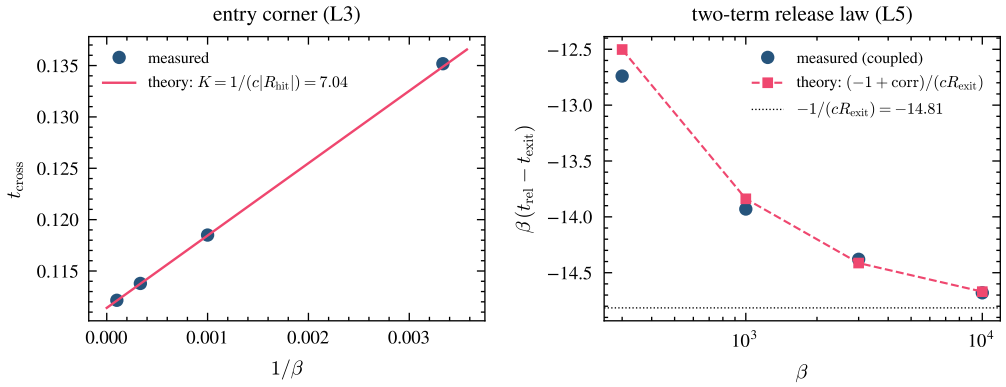


Figure 2: Predicted constants against measurements (deterministic runs; the uncertainty is discretization). Left: entry-corner crossing times against $1/\beta$; the line is the proved slope $K = 1/(c|R^{\rm hyb}(t_{\rm hit})|) = 7.04$ (Lemma 7), not a fit. Right: the two-term release law (Lemma 9, squares) against the coupled measurements (circles) over $\beta \in [3 \cdot 10^2, 10^4]$; the dotted line is the speed-independent first-order coefficient $-1/(cR^{\rm hyb}(t_{\rm exit})) = -14.81$. Every coupled run uses the explicit Euler step $dt = \min(10^{-5}, 0.02/\beta)$, so $dt = 10^{-5}$ at $\beta = 3 \cdot 10^2$ and 10^3, $6.7 \cdot 10^{-6}$ at $3 \cdot 10^3$ and $2 \cdot 10^{-6}$ at 10^4; Table 1 refines that step at $\beta = 10^4$. Script `t2-prova/figuras_draft.py`.

the tangential part of θ_A contributes $O(e^{-\beta\delta_1})$. Hence the supremum over this stage is $O_\delta(\log\beta/\beta)$, carried by the normal coordinate alone, which proves the sharpness asserted in Remark 3.

The terminal window and the release law are Lemma 9 step 5: the correction delay of Lemma 8(ii) is $O(\log\beta/\beta^2)$, the perturbation of the forcing shifts $t_{\rm exit}$ by $O(\log\beta/\beta^2)$ by Lemma 8(iii), and the area cut by the entry corner advances the exit by $-1/(cR^{\rm hyb}(t_{\rm exit})\beta)$. Summing the three gives (11), hence (5).

Finally Lemma 10 crosses the exit corner in time $O(\beta^{-1})$, selects the activating branch, and glues the two flows on $[t_{\rm release}, H]$ at rate $O(\log\beta/\beta)$. Taking the maximum of the four stage bounds gives (4). □

3.3 Numerical verification on one instance

The reference instance is $X = (0.5, 1.5)$, $Y = (0.5, 1.0)$, $\theta_A(0) = (-0.5, 0.27)$, $\theta_B(0) = (1.144, 0.261)$, giving $t_{\rm hit} = 0.1114$, $t_{\rm exit} = 1.1325$ and $\delta_1 = 0.508$. Every empirically measured constant is predicted by the proofs: the entry coefficient is $K_{\rm teo} = 7.04$ against 7.13 fitted (1.2%); the coupled two-term law $(-1+{\rm corr})/(cR_{\rm exit})$ reproduces the fine-grid coupled sweep $-12.74/-13.93/-14.38/-14.68$ to 1.9%/0.7%/0.2%/0.1% for $\beta = 3 \cdot 10^2 \ldots 10^4$, with the coefficient $1/(cR_{\rm exit}) = 14.815$ that the cancellation leaves (against -14.68 the leading coefficient alone is at 0.9%); the tangential drift is at machine zero. This subsection is one instance; Section 3.4 tests the same law on a sweep of drawn instances.

Table 1: A convergence test for the two-term law, not a table of measurements: the proved coefficient $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = -14.815$ is the limit, and the rows are successive refinements of the Euler step at $\beta = 10^4$ that must approach it. They do, monotonically, closing to 0.1% at the finest step; a law with the wrong constant would leave a floor here instead. The correction term of Lemma 9 is corr $= 0.00977$ at every step, so what is refining is the quadrature and not the law. At $dt = 10^{-5}$ the measurement is quantized in units of $\beta\, dt = 0.1$, which is the whole of that row's error. The sweep of Section 3.3 and Figure 2 uses $dt = \min(10^{-5}, 0.02/\beta)$, that is $dt = 2 \cdot 10^{-6}$ at $\beta = 10^4$.

dt	$\beta(t_{\mathrm{release}} - t_{\mathrm{exit}})$	deviation from -14.815
10^{-5}	-14.000	5.5%
$2 \cdot 10^{-6}$	-14.680	0.9%
10^{-6}	-14.760	0.4%
$5 \cdot 10^{-7}$	-14.805	0.1%

The measured coefficient is a discretization, not a fit, so it must be shown to converge. Table 1 is that test: it refines the Euler step at the stiffest point of the sweep, $\beta = 10^4$, keeping everything else fixed, and asks whether the measured offset approaches the *proved* limit -14.815 of (11) rather than some nearby number.
It converges to the proved coefficient: 0.1% at the finest step. The driver reuses `roda_acoplado` verbatim from `t2-prova/prova_L5_verificacao.py`, overriding only the step.

3.4 From one instance to a law

A single instance cannot distinguish a law from a coincidence of constants, so we ran the same test over drawn instances. The script `sweep_instancias.py` samples 600 triples $(X, Y, \theta(0))$ from a prior fixed in advance and documented in the script, integrates each with the same dt, the same integrator and the same $\beta = 10^4$ as the canonical run — the coupled driver is copied verbatim from `passo4_verificacao.py` — and compares the measured release offset $\beta(t_{\mathrm{release}} - t_{\mathrm{exit}})$ against the prediction $-1/(c\, R^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ formed from that instance's own constants. As a sanity check the reference instance of Section 3.3 is reproduced inside the sweep, at 2.9%.

Of the 600 draws, 57 satisfy (H1)–(H4). The exclusions are accounted for exactly: 340 have no death within the horizon, 162 fail (H2) because the area never returns to zero, and 41 die outside the detector's window. On the 57 admissible instances the resurrection occurs in 57/57 — the release always precedes the hybrid t_{exit} — and the relative error of the release coefficient against the instance's own $-1/(c\, R^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ has median 2.68%, ninetieth percentile 7.26% and maximum 12.01%. The predicted coefficients themselves range over $[-91.8, -4.5]$, a factor of 20, so the agreement is not the agreement of one number repeated. The worst quartile, those with |predicted| between 4.5 and 7.1, has median error 6.7%.

Two caveats belong with those numbers. First, the residual error is not centred: the median ratio of measured to predicted is 0.975 and 55 of the 57 instances have |measured| <

|predicted|. That systematic shortfall is the $o(\beta^{-1})$ remainder of Theorem 2 at a finite β, not sampling noise, and it is reported as a bias rather than absorbed into a tolerance. Second, the fraction 57/600 is a property of the prior and measures nothing about how large the set (H1)–(H2) is. The geometric number is the conditional one: of the 219 deaths that fall inside the detector's window, 57, that is 26%, satisfy (H2). The memory gate is a real and restrictive condition, not a formality that a generic trajectory satisfies.

3.5 Independence of the smoothing

The release criterion depends only on $\int R\,dt = 0$ and not on the sigmoid. For the erf class, $g'_\beta = \mathcal{N}(\beta z)$, the coupled system releases at the same t_{exit} with an $O(\beta^{-1})$ gap; this is verified numerically and not proved. GELU itself, $z\,\mathcal{N}(z)$, has a non-monotone derivative and is not a member of the family for any β; Section 6.6 shows that this is a distinct mechanism rather than a technicality.

4 Non-commutation and the plasticity bridge

Theorem 2 is the flow-first limit: one takes $\beta \to \infty$ of trajectories. The field-first order instead flows the pointwise-limit generalized gradient, and it freezes degenerate neurons forever. That branch is the rigorous one of Eberle, Jentzen, Riekert and Weiss [6, Theorem 3.3], whose selected solution is unique and has a degenerate set that is non-decreasing in time; their degenerate stratum is the fully-zero one rather than our functionally dead neuron, but the convention $\mathrm{ReLU}'(0) = 0$ and the no-revival mechanism are the same.

Proposition 11 (the two limit orders do not commute)**.** *Let L_∞ be the loss of the hard-ReLU network and let $\nabla^0 L_\infty$ be its pointwise-limit gradient, with the convention $\mathrm{ReLU}'(0) = 0$; let θ^{FF} solve $\dot\theta = -\nabla^0 L_\infty(\theta)$ from $\theta^{\mathrm{hyb}}(t_{\mathrm{hit}})$, which exists and is unique by [6, Theorem 3.3]. Assume (H1)–(H4). Then $\theta_A^{\mathrm{FF}} \equiv \theta_A(t_{\mathrm{hit}})$ on $[t_{\mathrm{hit}}, H]$, whereas θ^∞ of Definition 1 releases A at $t_{\mathrm{exit}} < H$. The two limits therefore disagree on $(t_{\mathrm{exit}}, H]$.*

Proof. The A-block of $\nabla^0 L_\infty$ at θ is $-\frac{1}{n}\sum_i \mathrm{res}_i\, \mathbf{1}_{\{z_{A,i}>0\}}(x_i, 1)$, which vanishes identically whenever every margin of A satisfies $z_{A,i} \le 0$. At t_{hit} the margin on datum j is zero and, by (H3), the remaining margins are $\le -\delta_1 < 0$, so the A-block vanishes and $\dot\theta_A^{\mathrm{FF}} = 0$. The margins of A are functions of θ_A alone, so the condition $z_{A,i} \le 0$ for all i is preserved as long as θ_A is constant, and the set of such θ is therefore forward invariant: θ_A^{FF} stays at $\theta_A(t_{\mathrm{hit}})$ on all of $[t_{\mathrm{hit}}, H]$. By Definition 1 and (H2), θ^∞ restarts the active flow at $t_{\mathrm{exit}} < H$ with $\dot z(t_{\mathrm{exit}}^+) = cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, so $z > 0$ immediately after t_{exit}. □

In machine-learning terms this non-commutation is the statement behind an empirical trend. Discrete ReLU units that die stay dead — an exact statement about a unit's *own* channel under a fixed input, which is the setting of Proposition 11 — and this is the setting in which the dormant-neuron and loss-of-plasticity phenomena are usually read [1, 2, 3], mitigated in practice by recycling dead units or by migrating to smooth activations; the smoothed flow resurrects them by the area law. The channel that reopens a smoothed

unit is $e^{\beta z}$, which closes as $\beta \to \infty$. Section 6.4 measures the resulting staircase directly. The qualification matters: once the input of the layer is itself moving, terminality is no longer implied, and in our transformer data 20% of injected hard-ReLU units return (20/100 against 0/100 with the upstream block frozen); Section 6.6 isolates that second channel.

5 Related work

Entry–exit and geometric singular perturbation. The release criterion $\int R\,dt = 0$ is the classical entry–exit function of slow–fast systems [5, Theorem 1.1], applied for instance to a neurotransmitter release model in [8], and Lemma 5 places our episode in the quadratic normal form of [5, equations (1.9)–(1.10)] with $\varepsilon = \beta^{-1}$. Our switching face is degenerate in a way that literature does not cover: it carries an identically zero field on the dead side. Generic piecewise-smooth exit theory [9, 10, 11] assumes a nonzero one-sided field, and the canonical Sotomayor–Teixeira regularization [12, 13] requires both one-sided normal components to be nonzero. The degeneracy is of the exponentially flat type treated by Kristiansen [14] with the blow-up method of Krupa and Szmolyan [15], but there the forcing is autonomous, whereas here the residual is generated by the very trajectory it gates. We are not aware of an analogue of that coupling in the entry–exit literature.

Gradient flows of ReLU networks. Existence, uniqueness and convergence rates for gradient flows of ReLU networks are established in [6], whose selected solution is the field-first branch of Proposition 11: once a neuron is degenerate it stays degenerate. Our result is the complementary, flow-first, selection, and the two disagree. The branch is fixed in practice by a convention of the differentiator, and that convention has a name: in the framework of Bolte and Pauwels [16] the derivative returned by automatic differentiation is a definable selection of a conservative field for the loss, and $\mathrm{ReLU}'(0) = 0$ is one such selection. Their Remark 13 also shows that the nondifferentiability locus $z = 0$ is reached with nonzero probability, so which branch the implementation picks is not a detail of measure zero: it is the choice that Proposition 11 shows the two limits disagree about.

Dormant neurons and loss of plasticity. Dead or dormant units are a measured pathology of trained networks: the dormant neuron phenomenon in deep reinforcement learning [1], the loss of plasticity under distribution shift [2], and its persistence in deep continual learning [3]. Death at initialization is a separate and better understood matter: Lu, Shin, Su and Karniadakis [17] prove born-dead results for ReLU networks at initialization and study the initializations that avoid them. We invoke them only for that regime; every confirmatory count in this paper conditions on *dynamical* deaths, units that die during training, and born-dead units are excluded by the instrument. The practical mitigations are recycling dead units and smoothing the activation. Theorem 2 and Proposition 11 say what the second mitigation buys: a channel of width $e^{\beta z}$ and a return time set by an area.

Dormancy is not death. The two objects should not be conflated. Dormancy in [1] is a normalized activation score falling below a threshold: a unit can be dormant and still have a nonzero column in J. Our object is the exact stratum on which the unit's column of J is identically zero on every datum. The sets are different, and the apparent tension with deep reinforcement learning — whose input is non-stationary by construction, and where dormancy nonetheless persists — dissolves once they are kept apart: a low-score unit need not be on the stratum, so nothing in this paper predicts that it must return. Our results separate two independent channels. The *own* channel requires smoothness of the activation, and its width is the factor $e^{\beta z}$, which the staircase of Figure 5 measures arm by arm; for hard ReLU that width is exactly zero. The *upstream* channel requires only that the layer's input move, and it operates for every activation we tested, hard ReLU included.

6 Experiments

The claims are tested by prediction, each with its null control, and the failed designs are reported alongside the successful ones. Every confirmatory analysis was frozen by a pre-registration commit before the run, and hypotheses that fell were archived rather than rewritten. The four instruments that carry these claims are stated as pseudocode in Section B. The commits are: pressure pilot `b48b527`; rate pilot `995a87c`; 4a.3 pre-registration `a8f8f8c`; 4a.3 confirmatory `5a961cb`; transformer pre-registration `1b45498`.

Statistics. Proportions are reported as a point estimate with a two-sided 95% Clopper–Pearson interval [18]; the variability captured is the Bernoulli count of returns among deaths within a single arm, with the denominator given in each cell. Event rates λ, dynamical deaths per run, are reported with a 95% percentile bootstrap interval over $B = 2 \cdot 10^4$ resamples clustered by seed, so that the variability captured is seed-to-seed and not episode-to-episode. Nothing is pooled across arms. We write T for the SGLD temperature, that is, the scale of the injected noise; $T = 0$ is deterministic gradient descent, and H remains the training horizon in optimizer steps.

6.1 The deterministic schedule is falsified under SGLD

Before any confirmatory claim we tested whether the deterministic entry–exit schedule survives noise. A reduced "race" model, pre-registered at commit `190d94a` after two adversarial-review rounds, predicts the capture (resurrection) probability of a dead unit from the mean residual schedule; the falsifying run of 2700 runs under SGLD [19] rejected it. The predicted-zero window was violated decisively: 139 late captures occurred where the model gives exactly zero, and the bin $[0, 0.1)$ saw 150 captures against 1.4 predicted (Poisson $z = +126$ against the model's own expected count, under the model's own independence assumption). The model is well calibrated where it says capture is likely, in the $(0.5, 0.9)$ bin, with $z = -0.26$; the entire error is on the "impossible" side. The theorem-part of the model held: in all nine cells the observed censoring of 2–7% matched the censoring rate that the

reduced model itself predicts, from a Brownian first-passage approximation to the margin inside the cone of the deterministic schedule. The failure is therefore of the hypothesis, a deterministic schedule, and not of the instrument.

A follow-up note records what happened next. The natural fix, a stochastic Ornstein–Uhlenbeck schedule with no free parameter, predicts p(capture) nearly identical to the deterministic one, with differences ≤ 0.013, and a late-capture rate of 0.1–1.5% against the $\sim 13.8\%$ observed, which is insufficient by an order of magnitude. That model was therefore refuted at the predictor stage and never run; the leading candidate became the back-reaction of the reactivating unit on the shared residual. The stochastic gate remains open and is out of scope for the theorem, whose regime is deterministic gradient flow.

6.2 Two instrument designs failed their own null control

This subsection is the most transferable part of the paper: any pipeline that counts rare unit-level events during training — dormancy dashboards included — can commit the two errors documented here, and the calibrated null that catches them is reusable as stated in Section B. The first attempt to see the law in a real MLP failed on the instrument, and the failure was caught by a pre-registered null (the first MLP stage, labelled `etapa-4a` in the repository; commit `6411ac8`, 600 runs, MLP with $m \in \{16, 64, 256\}$, SGLD with $\eta = 3 \cdot 10^{-4}$). The pure-diffusion arm *fired the calibrated null* ($m = 16$, $p = 0.002$) — the control that must stay silent did not; by the pre-registered rule the ruler is then biased and no scientific verdict on the substantive predictions may be claimed, so the run is invalid. The diagnosis is in Figure 3: the ruler was measuring boundary chatter rather than the physical death of the theorem. Of all episodes 59% lasted ≤ 5 steps, and hard ReLU "resurrected" at the same rate and with the same duration distribution as softplus at $T > 0$, which is impossible by gradient, since a dead ReLU unit has exactly zero gradient; these were noise threshold-crossings. The real physics lives at $T \to 0$: the only genuinely long episodes sit in the $T = 0$ arm, where just 2 of 461 episodes completed inside the 10^5-step horizon (672 and 1548 steps; 99.6% censored). The calibrated null (Algorithm 2, whose retention step is the clean-episode filter of that ruler) held its false-positive rate near α in i.i.d. worlds, but the real world has per-neuron heterogeneity.

A redesign, `etapa-4a.2`, was itself rejected at the design stage by the same process (commit `a413b19`): at $T = 10^{-5}$ the ReLU-to-softplus ratio stayed ≈ 0.95 after debounce, because gradient starvation at $\beta = 300$ masked the effect, and the pooled null broke α by 30–92$\times$ in heterogeneous and autocorrelated worlds. The inheritance was explicit: a hysteresis and dwell ruler, a per-neuron stratified null, and a move to the deterministic regime $T = 0$ with a much longer horizon. These negatives falsify neither the area law nor the toy calibration. They falsify two rulers, and they are what forced the working design.

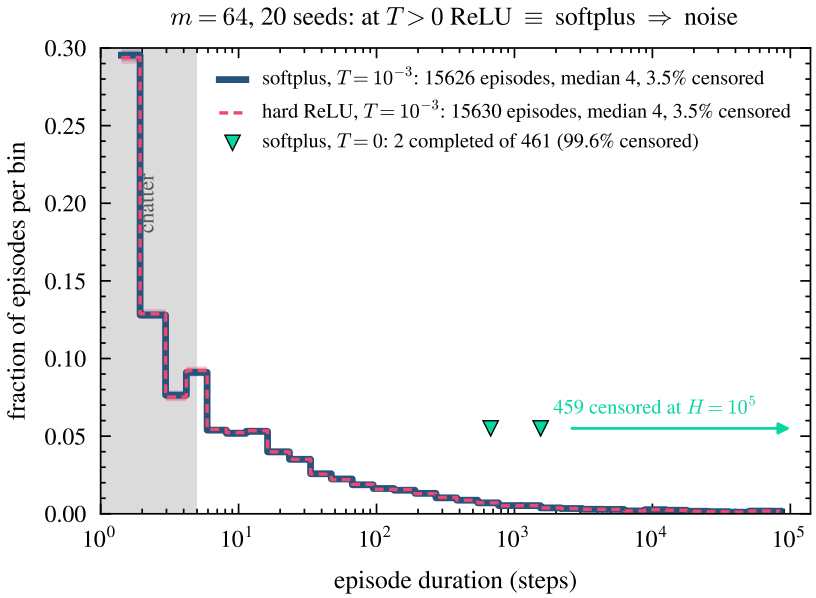


Figure 3: Why `etapa-4a` is an instrument negative (post-hoc diagnostic, $m = 64$, 20 seeds). Episode durations as the fraction of episodes per logarithmic bin; the bands are ± 1 standard error across the 20 seeds. At $T = 10^{-3}$ hard ReLU (dashed) lies on top of softplus (solid) — same median of 4 steps inside the shaded chatter band, same tail — which cannot be the zero-gradient theorem at work, since a dead ReLU unit has exactly zero gradient; these are noise threshold-crossings. The $T = 0$ arm completed only 2 of its 461 episodes within the same horizon, so those two are drawn as individual markers rather than as a density, and the remaining 99.6% are censored at $H = 10^5$; that is where the real physics sits and why the confirmatory moved to $T = 0$ with a longer horizon. Script `etapa4-mlp-populacao/figuras_diagnostico.py`.

6.3 A softplus unit dies and returns in a real MLP, and the area opens first

Moving to $T = 0$ and to a sub-parametrized regime made the mechanism visible. The regime is the *pressure* pilot: a plateau at ~ 0.15 on the rich target $y = \sin(2\pi x) + 0.3\sin(5\pi x)$ with $n = 16$, $\eta = 10^{-3}$ and $H = 2 \cdot 10^5$; 64 runs, commit `b48b527`, exploratory. Hard ReLU gave a structural zero: 0 resurrections out of 25 deaths across 16 seeds, so the frozen branch of Proposition 11 is exact in a real network at $T = 0$. Softplus resurrected by mechanism: at $\beta = 10$ a unit died at step 781 and returned at 139 179; at $\beta = 300$ one died at 3315 and returned at 152 977, having parked shallow enough that the $e^{\beta z}$ channel did not vanish. Both dynamical resurrections showed the ordered triple $t_{\text{death}} < t_{\text{area}} < t_{\text{res}}$, so the area law held 2/2. The gap between gate and return is the finite-β rise time, $\sim |z_{\text{park}}|/\text{drift}$, rather than the asymptotic $O(1/\beta)$ correction. The bottleneck was quantified as an event rate of

Table 2: Confirmatory 4a.3: 400 runs, 100 per arm, $T = 0$, $H = 5 \cdot 10^5$. λ is the number of dynamical deaths per run. $\hat{p}_{\text{death}}$ conditions on all dynamical deaths in the arm; $\hat{p}_{\text{elig}}$ conditions on the deaths that occur inside the eligibility window $[0, 307\,536]$, which is the only conditioning for which the return is observable within the horizon (for sp-β3 the eligible denominator is $N_{\text{elig}} = 30$ with 15 returns). Brackets are 95% Clopper–Pearson intervals, except the λ column, which carries a 95% bootstrap interval clustered by seed. ReLU is the validity control: any ground-truth return there would invalidate the run.

arm	dyn. deaths	λ/run [boot]	returns	$\hat{p}_{\text{death}}$	$\hat{p}_{\text{elig}}$
sp-β3	35	0.350 [0.220, 0.500]	16	0.457 [0.288, 0.634]	0.500 [0.313, 0.687]
sp-β10	27	0.270 [0.160, 0.390]	10	0.370 [0.194, 0.576]	0.385 [0.202, 0.594]
sp-β300	19	0.190 [0.100, 0.290]	4	0.211 [0.061, 0.456]	0.222 [0.064, 0.476]
ReLU	17	0.170 [0.090, 0.260]	0	0.000 [0.000, 0.195]	0.000 [0.000, 0.195]

~ 0.1 per run, and that number set the entire sizing of the confirmatory.

Figure 4 shows the $\beta = 10$ episode as it happened, in the three coordinates the theorem talks about: time, margin, accumulated area. The qualitative shape of Figure 1 — dive, sticking layer, debt, zero of the area, release — appears in a trained network with no scalar reduction and no smoothing of the data, only with the return lagging the gate by the finite-β rise time instead of by $O(1/\beta)$. The scope of this transfer is narrow: in this episode 68% of the lifetime elapses *after* the gate opens, so at $\beta = 10$ the asymptotic release formula is far from its regime. What transfers to the trained network is the *ordering* $t_{\text{death}} < t_{\text{area}} < t_{\text{res}}$ and the gate's necessity, not the $O(1/\beta)$ release-time expansion.

6.4 The area law reproduces under a pre-registered confirmatory

The confirmatory, `etapa-4a.3`, was pre-registered at commit `a8f8f8c` after two further adversarial rounds, with every design number derived programmatically from a 240-run rate pilot (commit `995a87c`); the run is commit `5a961cb`. Design grid: {sp-β3 (primary), sp-β10, sp-β300, ReLU} $\times\, m \in \{4, 6\} \times 50$ held-out seeds (1900–1949, not used in any pilot), for 400 runs at $T = 0$ and $H = 5 \cdot 10^5$. The eligibility window is $[0, 307\,536]$, that is $H - \tau$ with τ the $q_{0.9}$ return time of the pilot. The instrument is a single ground-truth clock with bit-exact parity to the pilot motor (Algorithm 1), read by the pre-registered decision procedure of Algorithms 4 and 5, and the area of the leader-datum residual is accumulated on that clock, so temporal ordering is structural and the only empirical failure mode of the area test is a return without area.

The verdicts (Tables 2 and 3 and Figure 5) are as follows.

- **C1, existence with rate, primary arm sp-β3: reproduces.** With $N_{\text{elig}} = 30$ against the floor $n^* = 10$, there are 15 eligible returns and $\hat{p}_{\text{elig}} = 0.500\,[0.313, 0.687]$, well above the kill threshold 0.076. Over all 35 dynamical deaths the same 16 returns give $\hat{p}_{\text{death}} = 0.457\,[0.288, 0.634]$; the two numbers are two conditionings of one count, not two estimates of one quantity.

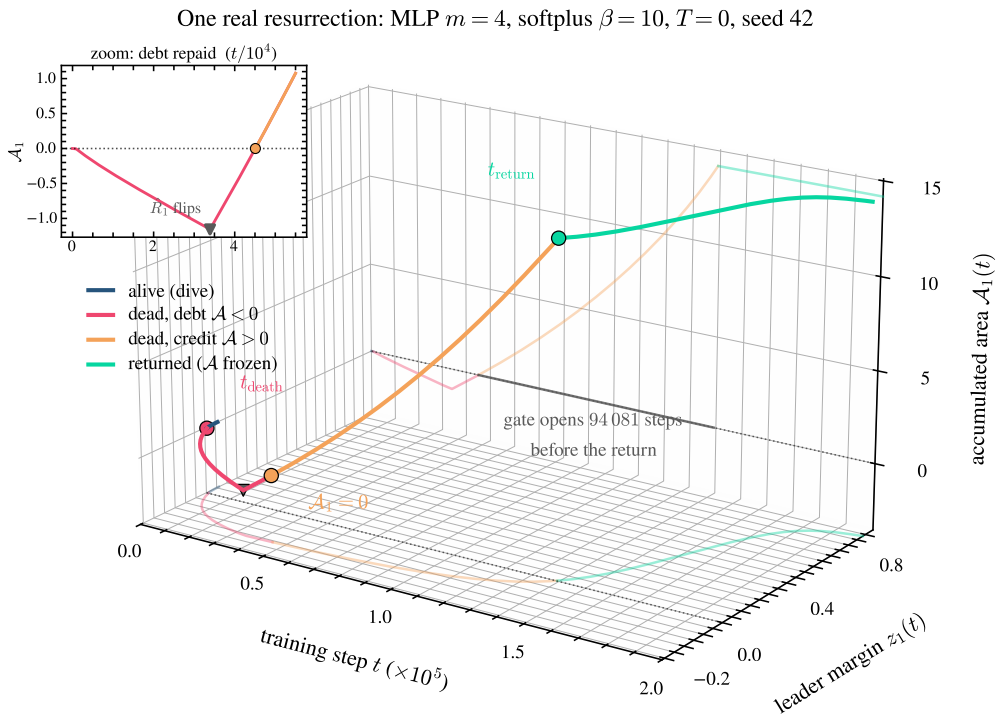


Figure 4: One real resurrection, drawn from the recorded trajectory of the run described above (pressure pilot, softplus $\beta = 10$, $m = 4$, $n = 16$, $\eta = 10^{-3}$, $T = 0$, $H = 2 \cdot 10^5$, seed 42, unit $k = 1$). The axes are the training step t, the leader margin $z_1(t)$ whose sign decides life or death, and the ruler's accumulated area $\mathcal{A}_1(t) = \eta \sum_{s=t_{\text{hit}}}^{t} \hat{R}_1(s)$; the pale curves on the two walls are the (t, z_1) and $(t, \mathcal{A}_1)$ projections. The unit dives and dies at $t = 781$ (t_{hit}). The area then runs into debt while the projected residual $\hat{R}_1$ is adverse, turns at $t = 33\,905$ where $\hat{R}_1$ changes sign (inset), and comes back to zero at $t = 45\,098$: the gate. The margin only then crawls up out of the sticking layer and crosses zero at $t = 139\,179$, so the gate opens 94 081 steps before the return — the ordering $t_{\text{hit}} < t_{\text{area}} < t_{\text{res}}$ of Theorem 2, with the lag being the finite-β rise time. The ruler stops accumulating at the return, which is why $\mathcal{A}_1$ is flat afterwards. Nothing is fitted or smoothed: the curve is the trajectory itself, sub-sampled every 40 steps, and the four marked steps are exact. Script `figura_trajetoria3d.py`, which re-runs the cell and refuses to plot unless the committed triple (781, 45 098, 139 179) is reproduced exactly.

Table 3: The pre-registered questions of 4a.3, with the condition that kills each one. All thresholds were derived programmatically from the rate pilot and frozen in `config_4a3.json` at commit `a8f8f8c`, before the run.

label	estimand	condition that kills	verdict
C1	p(ret | eligible), sp-$\beta 3$	CP upper < 0.076, $N_{\text{elig}} \geq 10$	reproduces
C2	returns with area first, sp-$\beta 3$	CP lower < 0.5, ≥ 6 returns	sustains
C3	rise gap against park depth	none (estimation)	not aggregated
C4	p(ret | death) against β	none (estimation)	monotone in β
V	returns in the ReLU arm	any single return	clean, 0/17

- **C2, the area law, sp-$\beta 3$: sustains.** The leader-drive area opens before the return in 13/16 dynamical resurrections, with lower Clopper–Pearson bound $0.544 > 0.5$ on ≥ 6 returns. The ordering itself is partly structural: the clock accumulates the area only on dead steps and closes at the return, so $t_{\text{area}} < t_{\text{res}}$ cannot fail whenever the area opens at all (Section B.1). The empirical content is that the area in fact turned positive in 13/16 cases; the 3 returns without area are real and are counted against the law: they are returns driven off the leader datum, which the single-datum hypothesis (H3) does not cover; a per-datum area $\mathcal{A}_i$ is the natural extension. What this instrument can say about the gate stops there, as the post-hoc aggregation of Section D makes explicit.

- **C3, the rise gap: estimation, not aggregated.** C3 was pre-registered as an estimation-only quantity with no kill condition, relating the gap $t_{\text{res}} - t_{\text{area}}$ to the park depth z_{park} and to the channel $\sigma(\beta z_{\text{park}})$. Those quantities are recorded per episode by the instrument, but the frozen analysis script contains no aggregate for C3. A post-hoc aggregation, produced after the confirmatory data were seen and labelled as such, is reported in Section D: it finds no evidence that the opening of the area predicts the return time beyond the onset of the episode. The episode's mean drive was never persisted by the instrument, which is why the question cannot be pushed further; persisting the drive series is inherited as a requirement for future experiments.

- **C4, channel staircase: estimation, monotone in β.** $\hat{p}_{\text{death}}$ falls $0.457 > 0.370 > 0.211 > 0.000$ from $\beta = 3$ to hard ReLU, with the intervals of Table 2. The pairwise intervals overlap; the pre-registered protocol declared C4 an estimation with no kill condition, so no trend test was frozen in advance. Conditioning on death when β also moves the death rate (35/27/19/17 dynamical deaths across the four arms) introduces a collider bias in $\hat{p}_{\text{death}}$; C4 was already estimation-only, and this note makes the caveat explicit. As *post-hoc supporting analysis* we report the Cochran–Armitage trend test across the four ordered arms: $z = -3.49$, $p = 4.9 \cdot 10^{-4}$ (pairwise Fisher, $\beta = 3$ vs. ReLU: $p = 8.1 \cdot 10^{-4}$; $\beta = 3$ vs. $\beta = 300$ alone: $p = 0.086$, not significant — the trend test, not pairwise comparison, is the appropriate instrument for an ordered dose–response). Likewise for C2, a post-hoc exact binomial test of 13/16 against 0.5 gives one-sided $p = 0.011$ (two-sided 0.021); the pre-registered criterion was the

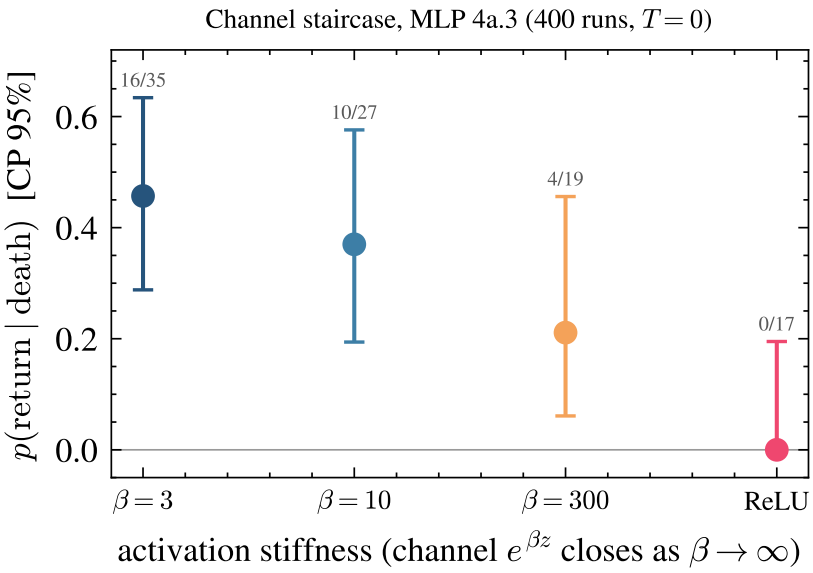


Figure 5: The channel staircase (C4, confirmatory 4a.3, 400 runs, $T = 0$). $\hat{p}_{\text{death}}$ is monotone in the activation stiffness and reaches a structural zero at hard ReLU, where the gradient vanishes identically; the smooth channel $e^{\beta z}$ closes as $\beta \to \infty$. The fractions above each point are returns over deaths and the error bars are 95% Clopper–Pearson intervals. Script `figuras_paper.py`.

Clopper–Pearson bound, which these tests corroborate but did not replace. The activation smoothness controls the return channel, and $\beta = 3$, the stiffness scale of the GELU family, is the arm with the highest return probability. This is the direct bridge to the plasticity practice of Section 4.

- **Validity: clean.** ReLU gave 0/17 dynamical returns, which is Proposition 11 measured with $N = 100$ runs.

The limits of the confirmatory are $T = 0$, one small architecture, a one-dimensional rich target, and $m \in \{4, 6\}$. The stochastic regime is a paper of its own (Section 6.1). The kill condition of C1 is intentionally near-unreachable at this N, a consequence of the frozen rule that halves the pilot lower bound, so in practice C2 is the deciding test; this was signalled in the pre-registration rather than discovered afterwards.

6.5 The theorem class transfers to a transformer feed-forward block

The same question in a one-layer transformer, for the feed-forward block over an attended token, does not reduce to the MLP. The scoping run (exploratory, commit `d4e4c35`) exposed two mechanisms that reshaped the design, a GELU sign-flip barrier and an upstream resurrection channel; both are stated in Section 6.6. The confirmatory therefore tests the

theorem class, softplus-β, on the block, and treats the barrier and the upstream channel as estimands of their own.

A rate pilot of 160 runs (commit `2dba27d`) then fixed the design. Natural FFN death is ≈ 0, since units with $d = 16$ input do not die on their own, so the confirmatory is necessarily injection-conditional; the non-stationary channel chatters, forcing the hysteresis instrument of Algorithm 3, with $\delta_h \approx 0.0476$ calibrated from data; and the return time is slow, $\tau \approx 60\,000$–$66\,000$ steps, forcing $H = 2 \cdot 10^5$. The upstream channel was already clean in the pilot: the difference between training and frozen attention was $+6/+6/+6/+8$ returns, and ReLU went from 0 to 8 with own-move 0.

The pre-registered protocol. A confirmatory run for the transformer was pre-registered at commit `1b45498`: 800 runs over {sp-β3, sp-β10, GELU, ReLU} $\times$ {attn-train, attn-frozen}$\times$ injection $\{-0.3, -2.0\} \times 50$ seeds 2000–2049, with $H = 2 \cdot 10^5$ and $T = 0$. The grid completed with 800/800 runs (manifest `manifest-e5.json`), with no change to the estimands. The decision procedure is Algorithms 4 and 5. The five pre-registered estimands E-A to E-E are stated verbatim in Section C, so that the protocol is on record ahead of the data.

The verdicts. Table 4 carries the frozen analysis, one row per estimand: the validity control is clean, the theorem class transfers through the unit's own channel, the upstream channel reproduces in all four arms, the β ordering of the MLP staircase repeats inside the block, and the GELU barrier is left consistent-with but confounded with depth. Section C keeps each estimand as it was frozen, with its result attached; Figure 6 shows the whole grid at a glance.

6.6 GELU sits outside the class and the moving input opens a second channel

The scoping run of the block (commit `d4e4c35`) produced two facts that are mechanisms in their own right, and that the confirmatory above carries as E-D and E-B:

- **S1, a GELU sign-flip barrier.** GELU$'(z)$ changes sign at $z^* = -0.7518$, verified analytically: for $z < z^*$ it is negative, so a deep-dead unit is pushed away from the return by its own gradient. Self-resurrection is possible only in the shallow band $(-0.752, 0)$; Figure 7 draws the barrier and both injection depths. The MLP's blind area port, which omits g', gets the sign wrong on 12/12 GELU episodes; the correct instrument uses the slope-weighted area $\mathcal{A}_{\text{drive}}$. GELU is therefore genuinely outside the monotone-sigmoid class of Theorem 2. There is a prior point: GELU has no dead stratum to begin with. A unit at margin -0.3 emits GELU$(-0.3) = -0.115 \neq 0$ and keeps contributing to the residual, so $\{z \leq 0$ on every datum$\}$ is not a singular stratum for GELU but a region of weak, sign-inverted coupling. The sign-flip barrier is the symptom that the definition of death does not apply, and it is consistent with GELU being the most revivable arm we measured (shallow frozen 0.580); we note that

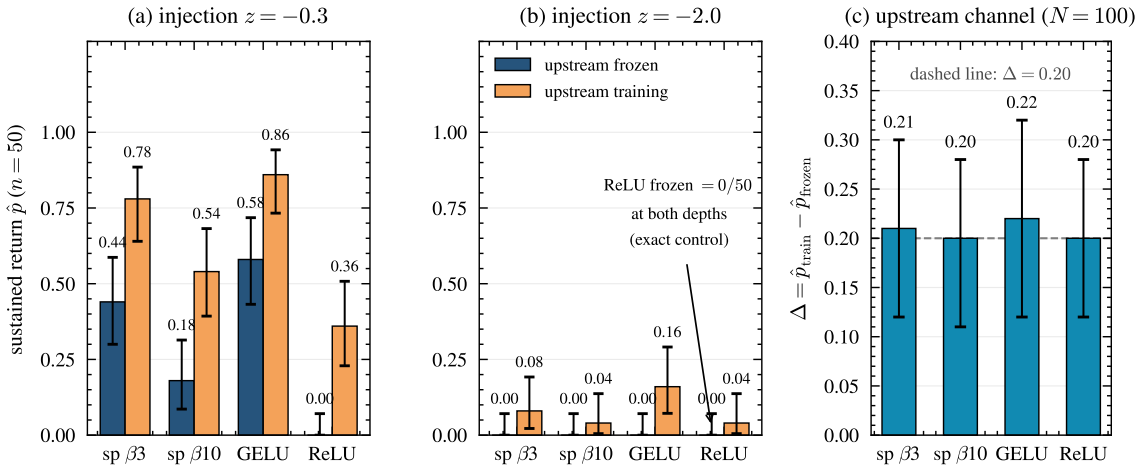


Figure 6: The confirmatory grid on the transformer feed-forward block (800 runs; 50 per activation-by-depth-by-regime cell; hysteresis $\delta_h = 0.0476$, horizon $H = 2 \cdot 10^5$). (a), (b) Sustained-return rate $\hat{p}$ with 95% Clopper–Pearson intervals at shallow ($z = -0.3$) and deep ($z = -2.0$) injection. (c) The upstream channel: $\Delta = \hat{p}_{\text{train}} - \hat{p}_{\text{frozen}}$ pooled over both depths (100 runs per arm, 95% bootstrap-by-seed intervals). Three readings: Δ is uniform at ≈ 0.20 across all four activations, so the upstream channel is not an artefact of any one nonlinearity; relu-frozen is exactly 0/50 at both depths, the designed negative control; and shallow GELU is the highest cell (0.580 frozen, 0.860 training), consistent with GELU having no dead stratum (Figure 7). Script `figura_grade_e5.py`.

against sp-β3-frozen (29/50 against 22/50) the difference is not statistically separable, two-sided Fisher exact test, $p = 0.23$.

- **S2, an upstream channel.** The input of the block, $h = e + \text{attn}(e)$, is non-stationary, so a hard-ReLU unit whose own gradient is identically zero can still revive because h moves. With attention frozen the control is exact: the margin change is 0.000 and the own-move is 0. With attention training the own-move is still 0, yet the margin climbs by +0.69 on average and by up to +1.32. The MLP statement "dead ReLU is terminal" holds only when the input is fixed; the transformer adds a channel the MLP does not have.

Neither mechanism is a technicality of the instrument. The barrier says that the MLP mechanism does not port unchanged to a GELU block, since the sign-flip obstructs self-resurrection below z^*; the upstream channel says that terminality of a discretely dead unit is a statement about that unit's own channel under a stationary input, and nothing more. In the confirmatory the barrier's terminality could not be separated from injection depth, and it is reported that way (E-D of Table 4); the upstream channel is established in all four arms (E-B).

Table 4: Transformer confirmatory (`etapa-5`): 800 runs, $T = 0$, $H = 2 \cdot 10^5$, pre-registered at commit `1b45498`. One row per frozen estimand; the wording of each target is the pre-registration's, reproduced verbatim in Section C. Brackets are 95% Clopper–Pearson intervals for proportions and 95% bootstrap intervals for the differences Δ. Denominators are 50 runs per arm-by-depth cell and 100 per arm when the two depths are pooled.

estimand	pre-registered target	result	interval or test
E-A, validity	no sustained return in the relu-frozen arm	valid: 0/100 sustained returns, both depths	— (a single return would have invalidated the run)
E-B, upstream channel	$\Delta = p_{\text{sust}}(\text{train}) - p_{\text{sust}}(\text{frozen})$ with positive bootstrap lower bound, per arm	reproduced in all four arms: 0.210 (sp-β3), 0.200 (sp-β10), 0.220 (GELU), 0.200 (ReLU)	$[0.120, 0.300]$; $[0.110, 0.280]$; $[0.120, 0.320]$; $[0.120, 0.280]$. ReLU contrast 20/100 against 0/100, two-sided Fisher exact test, $p = 6.6 \cdot 10^{-7}$
E-C, transfer of the theorem class	at least $n^* = 6$ sustained returns, sp-β3 frozen, shallow injection	transferred: 22/50 sustained returns, 3.7× the gate; the leader-drive area opened before the return in 21/22	$\hat{p} = 0.440\,[0.300, 0.587]$; gate $n^* = 6$
E-D, GELU barrier	shallow injection far above deep in the own channel, against a depth-matched control	consistent with the barrier, confounded with depth (the frozen three-way label): GELU 29/50 shallow against 0/50 deep, but sp-β3 deep also gives 0/50	S1 signature, $\text{sign}(\mathcal{A}_{\text{drive}}) \neq \text{sign}(\mathcal{A}_{\text{mlp}})$, in 92% of deep against 4% of shallow episodes
E-E, β staircase	estimation across $\beta \in \{3, 10\}$, frozen and training, no kill condition	$\beta = 3$ above $\beta = 10$ in both factors: frozen 0.220 against 0.090, training 0.430 against 0.290; GELU is the most revivable arm	GELU shallow frozen $0.580\,[0.432, 0.718]$; with the upstream channel $0.860\,[0.733, 0.942]$

The deep zero is what the channel law predicts. The own channel of sp-β3 is $\sigma(3z)$, which is 0.289 at the shallow injection $z = -0.3$ and 0.00247 at the deep one $z = -2.0$: a ratio of about 117. The return time measured in the shallow cells is $\tau \approx 60\,000$–$66\,000$ steps, so scaling it by that ratio projects a deep return at roughly $7 \cdot 10^6$ steps, about 35 times the horizon $H = 2 \cdot 10^5$. The 0/50 of sp-β3 frozen at $z = -2.0$ is therefore predicted by the channel law itself, at this horizon, and is not a failure of the transfer. It also disarms part of the E-D confound: the deep suppression of softplus is starvation of a quantitatively predicted size, so the depth-matched control gives 0/50 for a reason that has nothing to do with the GELU barrier, and the two mechanisms remain separable only by the depth sweep around $z^* = -0.752$ that Section C specifies.

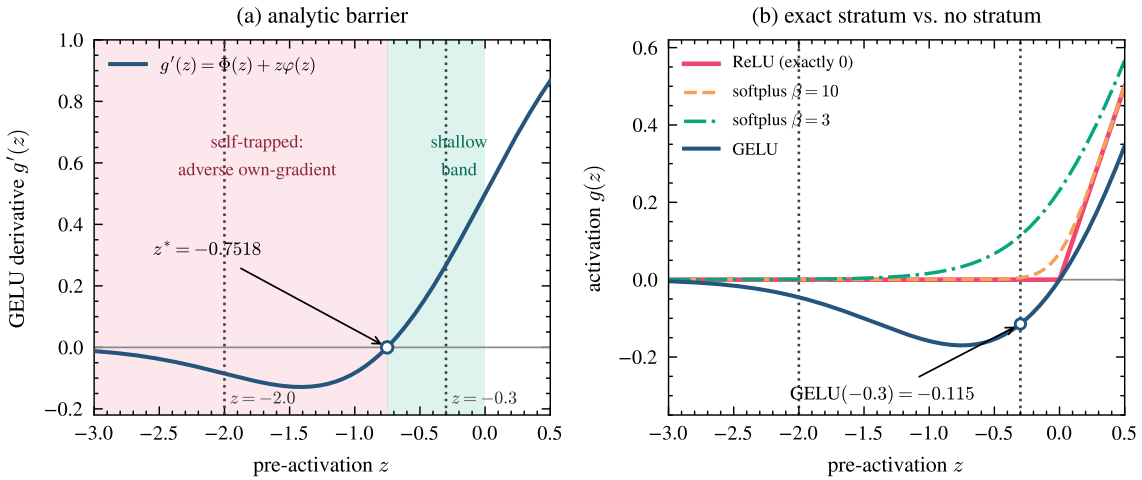


Figure 7: The analytic barrier and the non-stratum, both exact. Left: the GELU derivative $g'(z) = \Phi(z) + z\varphi(z)$ changes sign at $z^* = -0.7518$; below z^* (shaded) the unit's own gradient is adverse and the region is self-trapping, while on the shallow band $(z^*, 0)$ the own gradient still points back toward activity. The two injection depths of the confirmatory, $z = -0.3$ and $z = -2.0$, sit on opposite sides of the barrier. Right: the activations themselves. ReLU vanishes identically for $z < 0$, an exact dead stratum; softplus-β is exponentially small but strictly positive; GELU is strictly negative, with GELU$(-0.3) = -0.115$, so for GELU there is no dead stratum at all and the deep-injection suppression of Figure 6 must come from the sign of g', not from a loss of rank. Script `figura_barreira_gelu.py`.

6.7 Deviations from the pre-registered protocol

(i) None for 4a.3: the analysis frozen at `a8f8f8c` was run unchanged at `5a961cb`, and the validity control passed. (ii) The eligibility window $[0, 307\,536]$, the kill threshold 0.076, the floor $n^* = 10$ for C1 and the floor of ≥ 6 returns for C2 were all derived programmatically from the rate pilot (`995a87c`) and frozen in `config_4a3.json` before the run; none was chosen after seeing the confirmatory data. (iii) C3 was pre-registered as estimation-only; the frozen script reports no aggregate for it, and the post-hoc aggregation of Section D is labelled as such (Section 6.4). (iv) For the transformer, the pre-registered package was revised once before freezing: an earlier candidate was rejected in adversarial review for a confound between injection depth and the GELU barrier, and was replaced by the package of commit `1b45498`, which adds a depth-matched control. No run was executed under the rejected package.

7 Limitations

(i) The theorem is proved for the minimal configuration: one unit, one active datum at a time, a fixed readout and a one-dimensional input. Every constant is verified on a reference instance and the release law holds on 57/57 of the admissible instances of a 600-draw sweep (Section 3.4), with a residual bias, 55/57 measured offsets smaller in modulus than predicted, that we read as the $o(\beta^{-1})$ remainder at finite β. The experiments train the readout v_k, so the transfer from the proved object to the measured one is empirical; we are explicit about this rather than treating the two as the same object. **(ii)** The regime is $T = 0$ deterministic gradient flow throughout the theorem and the confirmatory; the stochastic gate is open and, where probed, falsifies the deterministic schedule (Section 6.1), since late captures recur past the deterministic window at a rate the reduced stochastic model under-predicts, leaving the reactivating unit's back-reaction as the leading candidate driver. **(iii)** Scale: the confirmatories are small networks ($m \leq 6$ MLP units, $m_{\text{ffn}} = 4$ block units) with rich one-dimensional targets, chosen so that a single unit can be tracked with a ground-truth clock. **(iv)** GELU is outside the monotone class; its own-channel revival is blocked below $z^* = -0.752$, and in the confirmatory the barrier could not be separated from depth (the pre-registered three-way label), because at the deep injection $z = -2.0$ the softplus own channel is starved to 0.9% of its shallow value — a factor of about 117, which projects the deep return past 35 horizons (Section 6.6) — while GELU$'$ is adverse, so both mechanisms predict the same zero at this horizon; the separating design, a sweep of injection depths around $z^* = -0.752$, is stated in Section C and left as future work. The upstream channel is established; the barrier's terminality remains open. **(v)** The cone apex, of corank ≥ 2, where weight decay produces genuine sliding toward the most singular stratum, is untouched. **(vi)** The independence of the law from the smoothing is proved for softplus and verified numerically for erf; it is not proved for the whole monotone-sigmoid class.

8 Conclusion

We derived an entry–exit law for the death and resurrection of a smoothed unit, verified each of its constants on a two-neuron instance, and tested the release law over a sweep of 600 drawn instances, where it holds on all 57 that satisfy the hypotheses, with a median relative error of 2.7%. We measured the law in a small MLP under a pre-registered protocol: resurrection occurs, the residual area opens before the return in 13 of 16 cases, hard ReLU gives a structural zero, and the return probability is monotone in the activation stiffness. In a pre-registered transformer confirmatory the theorem class transferred to the feed-forward block through the own channel (22/50, area before the return in 21/22), a second upstream channel reproduced in all four arms, and GELU's sign-changing derivative left its signature in 92% of deep episodes while its terminality remained confounded with depth — the pre-registered three-way label, reported as frozen.

What this is, and is not. The value of the law is understanding: it replaces "dead units are terminal" with a mechanism — a gate with memory — in the regime where the

mechanism is provable. It is not a training tool. The asymptotics live on the channel $e^{\beta z}$ being exponentially small, and one would expect an adaptive optimizer that divides by the running RMS of the gradient to rescale precisely that smallness. An exploratory pilot on the toy instances (13 instances, `e7-adam-portao` in the repository) measured this expectation instead of assuming it, and it fails in an instructive way: under Adam the gate survives, parked deeper by the optimizer's floor — the release moves to where the channel crosses ϵ, empirically $\beta z_{\text{park}} \approx -\ln(1/\epsilon) + \text{const}$ — and the area law itself reappears as ϵ grows, with relative error 0.24% at $\epsilon = 1$ against 0.017% for plain gradient descent. This is a pilot, not a confirmatory, and momentum-rich production regimes remain unmeasured. What does transfer is smaller and sharper: the sign structure of the activation derivative, which is an exact fact about GELU (Section 6.6) and, with the zero at $z^* = -1.2785$ instead of -0.7518, about SiLU; the calibrated-null instrument of Section 6.2, which any measurement of rare events in training can reuse; and the folklore correction that the terminality of discrete death is a statement about stationary input. The cheapest consequential experiment this paper points at is not a scaling of the theorem but a histogram: the mass of feed-forward pre-activations below z^* in production-scale models, and whether that mass is sticky across training.

9 Reproducibility

All code, configs, per-run manifests and aggregate JSON needed to reproduce every number and figure in this note are in a single Git repository. One policy governs the raw data, in both confirmatories: the per-episode trace files (`results-4a3/*.jsonl.gz`, `results-e5/*.jsonl.gz`) are kept out of version control, while the manifests that checksum them (`results-4a3/manifest-4a3.json`, `results-e5/manifest-e5.json`) are versioned. The traces are byte-reproducible from the pinned configs (deterministic single-threaded CPU runs), so the versioned SHA256 list is what makes a regenerated file verifiable. That repository is private at the time of writing, so the commit hashes below, while they resolve in it and fix the state of every experiment, are not independently checkable today; public release under the MIT licence is a precondition for submission. All runs are deterministic on CPU with per-run manifests (SHA256, git hash, config), and pre-registration commits precede every confirmatory run in the repository history. No external data were used; every dataset in this note is generated by the scripts listed below.

- **Theorem constants and figures.** `06-notas/t2-prova/`: lemma verifications `passo0--6.py` and `prova_L1--L6.py`; figures via `figuras_draft.py`.
- **MLP mechanism (pilot).** `etapa4-mlp-populacao/pilotos-4a2/pressao/` (the *pressure* regime: sub-parametrized, plateau ~ 0.15); commit `b48b527`. The trajectory of Figure 4 is re-run and plotted by `06-notas/paper/figura_trajetoria3d.py`, which asserts the committed triple before drawing anything; the diagnostic of Figure 3 is `etapa4-mlp-populacao/figuras_diagnostico.py`.

- **MLP confirmatory (4a.3).** `.../pressao/taxa/` (the event-rate pilot and the confirmatory built on it): pre-registration `a8f8f8c`, run `5a961cb`. `python run_4a3.py --workers 4` then `python analise_4a3.py`, which reads `results-4a3/` and `config_4a3.json`. Rate pilot `995a87c`. Staircase figure: `06-notas/paper/figuras_paper.py`.
- **Transformer (`etapa-5`).** Directory `05-experimentos/etapa5-transformer/`; scoping `d4e4c35`, rate pilot `2dba27d`, confirmatory pre-registration `1b45498`. `python run_e5.py --workers 4` then `python analise_e5.py`.
- **Negatives.** `05-experimentos/NEGATIVOS.md` (the register of negative results) and the `etapa-4a` note record the invalidated instrument (commit `6411ac8`) and the rejected redesign (`a413b19`); the falsified schedule is `p5-relu-morte-v2/` (commit `190d94a`).

Compute. Every run is single-threaded CPU on an Intel Core i9-9900K, with 4 concurrent workers through Python `multiprocessing` at `nice 19` and no GPU; the only dependency is `numpy` (Python 3.14, numpy 2.4). The 4a.3 confirmatory, 400 runs at $H = 5 \cdot 10^5$, took 4.9 h wall-clock, that is 19.5 CPU-hours at 175 s per run. The transformer confirmatory, 800 runs at $H = 2 \cdot 10^5$, took 13.9 h wall-clock, which includes one stall of the runner and its resumption. The preliminary and discarded runs, namely `etapa-4a` (600 runs), the `4a.2` redesign, the pressure pilot (64 runs) and the two rate pilots (240 and 160 runs), are of the same order again and are not otherwise reported.

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Data availability

Everything needed to reproduce every number and figure — code, configs, per-run manifests and aggregate JSON — is held at the commits cited in Section 9, under the trace-file policy stated there. The repository is not public at the time of writing; no external data were used.

Declaration of LLM usage

Large language models were used throughout the work: to draft and copy-edit the text, to run the adversarial reviews, to write the experiment and verification code, and to work out the derivations of Section A. The hypotheses, the pre-registered kill conditions, the decision to declare a design or a hypothesis dead, and the reading of the cited sources are the author's.

A review that only re-read the derivation would be circular, so the checks that carry weight here are the ones that do not read it. Each lemma is re-derived symbolically from scratch by a committed script (SymPy), which recomputes the identity rather than confirming the prose; the convergence table Table 1 refines the integrator until the measured coefficient meets the proved one; and the sweep of Section 3.4 confronts the release law with 600 independently drawn trajectories. These routes test the law against the dynamics, not against the text that describes it, and any of them can fail while the prose stands. The central identity, Lemma 5, is in addition a computation of about ten lines and is presented in that form so that a reader can redo it by hand.

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A Proofs of the six lemmas

Each proof closes with the outcome of its committed executable verification. The scripts are symbolic where an identity is claimed and numerical where a rate or a constant is claimed; they are listed in Section 9. Throughout, $\sigma(s) = 1/(1+e^{-s})$, $w = e^{\beta z}$, $u = 1/w$

and $\Psi(u) = \log u + u$, and we use freely the elementary identities $\sigma(\beta z) = 1/(1+u)$, $\beta\, g_\beta(z) = \log(1+w)$ and $(1-\sigma(s))/\sigma(s) = e^{-s}$, all of which are verified symbolically in `passo0_verificacao.py`.

A.1 Normal form and additive quadrature

Lemma 5 (normal form and additive quadrature). *Let $w = e^{\beta z}$ and $u = 1/w$, and set $\Psi(u) = \log u + u$. In fast time $\tau = \beta t$ the margin obeys*

$$\frac{dw}{d\tau} = c\, R(t)\, \frac{w^2}{1+w}\,(1+E_\beta), \qquad |E_\beta| \le \frac{\log(1+w)}{\beta|R|}, \tag{6}$$

the quadratic normal form of De Maesschalck and Schecter [5, equations (1.9)–(1.10)] with $\varepsilon = \beta^{-1}$, and one has the exact additive identity

$$\frac{d\Psi(u)}{dt} = -c\,\beta\, R(t) + c\log(1+w) + \mathcal{E}_{(\mathrm{H3})}(t), \qquad \big|\mathcal{E}_{(\mathrm{H3})}\big| \le C_3\,\beta\, e^{-\beta\delta_1}. \tag{7}$$

Under (H1)–(H2) the additive correction satisfies $\int_{t_{\mathrm{hit}}}^{t_{\mathrm{release}}} c\log(1+w)\,ds = O(\log\beta/\beta)$.

Proof of Lemma 5. Step 1 (margin equation, exact under (H3)). $\dot z = (x_j, 1)\cdot\dot\theta_A$, and the field on θ_A is $-\frac{1}{n}\sum_i \mathrm{res}_i\,\sigma(\beta z_i)(x_i,1)$. The term $i=j$ contributes $-c\,(g_\beta(z)-R)\,\sigma(\beta z)$, because $\mathrm{res}_j = g_\beta(z) - R$ by the definition (2) of R; the terms $i \ne j$ are bounded by $C_3' e^{-\beta\delta_1}$ by the scalar reduction of Lemma 10. Hence $\dot z = -c(g_\beta(z)-R)\sigma(\beta z) + O(e^{-\beta\delta_1})$, which in the gauge w and fast time $\tau = \beta t$ is (6).

Step 2 (exact chain rule). $\dot u = -\beta u\,\dot z$ and $\Psi'(u) = (1+u)/u$; together with $\sigma(\beta z) = 1/(1+u)$,

$$\frac{d\Psi}{dt} = \frac{1+u}{u}\cdot(-\beta u)\cdot\Big[-c\big(g_\beta(z)-R\big)\tfrac{1}{1+u}\Big] = c\,\beta\,\big(g_\beta(z)-R\big).$$

Step 3 (additive identity). Since $\beta\, g_\beta(z) = \log(1+e^{\beta z}) = \log(1+w)$, substitution gives $d\Psi/dt = -c\beta R + c\log(1+w)$, plus the (H3) error of Step 1 carried through the same factors, which is bounded by $C_3\beta e^{-\beta\delta_1}$. This is (7). No division by R occurs, so the interior zero of R forced by (H1)–(H2) through the intermediate value theorem is harmless.

Step 4 (bound on the correction, in three windows). On the episode $w = 1/u$ with $u = c\beta|\mathcal{A}(t)|(1+o(1))$ away from neighbourhoods of the endpoints (Lemma 8(i)). In the bulk, where $|\mathcal{A}| \ge a_0(\delta)$, one has $\log(1+w) \le w \le 1/(c\beta a_0)$, so that stretch contributes at most $T_{\mathrm{ep}}/(\beta a_0) = O(\beta^{-1})$, where $T_{\mathrm{ep}} = t_{\mathrm{exit}} - t_{\mathrm{hit}}$ is the duration of the episode, not the training horizon H: the bulk stretch lies inside the episode by construction. Near t_{hit} and t_{exit} the function $\mathcal{A}$ has transversal zeros, with derivatives $cR(t_{\mathrm{hit}}) \ne 0$ and $cR(t_{\mathrm{exit}}) \ne 0$, so $|\mathcal{A}(t)| \ge \kappa|t - t_*|$ locally and

$$\int \log(1+w) \;\lesssim\; \int_{1/\beta}^{\epsilon} \frac{ds}{c\beta\kappa s} = \frac{\log(\beta\epsilon)}{c\beta\kappa};$$

on the $O(\beta^{-1})$-wide stretches glued to the endpoints one has $w = O(1)$ and $\log(1+w) \le \log 2 + O(1)$, contributing $O(\beta^{-1})$. Summing the three windows gives $O(\log\beta/\beta)$ with constant $C_4 = 2/\big(c\min(|R(t_{\rm hit})|, R(t_{\rm exit}))\big) + T_{\rm ep}/a_0$. □

Remark 12 (consequence used downstream). Integrating (7) gives $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + O(\log\beta/\beta)$ uniformly on the episode. This is the form that sustains the depth $u = c\beta|\mathcal{A}|(1+o(1))$ on compacts and the release at $t_{\rm exit} + O(\beta^{-1})$, and it is what Lemmas 8 and 9 consume.

Verification. `prova_L1_verificacao.py` confirms the additive identity symbolically, and the integral correction with the normalized metric is stable in $[8.2, 10.6]$, below the leading theoretical term $2/\big(cR^{\rm hyb}(t_{\rm exit})\big) = 2/(0.625 \times 0.108) = 29.6$, for $\beta \in [3\cdot 10^2, 10^4]$. The full theoretical constant adds the bulk term: with $T_{\rm ep} = t_{\rm exit} - t_{\rm hit} = 1.1325 - 0.1114 = 1.021$ and $a_0 \approx 0.02$ in the instance, $C_4 = 29.6 + 1.021/0.02 = 80.7$. Using the training horizon $H = 1.7$ in place of $T_{\rm ep}$ would give 114.6 instead; the bound of Lemma 5 is over the episode, so $T_{\rm ep}$ is the correct length.

A.2 The B-subsystem and the feedback

Lemma 6 (B-subsystem and feedback). *Under (H4) the B-field is affine with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, where G_B is the Gram matrix of the active data, and $R^{\rm hyb}$ is affine in θ_B. Integral Grönwall consuming the bound of Lemma 5 gives, conditionally on the layer being maintained (supplied by Lemma 9),*

$$\sup\|\theta_B^{\beta} - \theta_B^{\rm hyb}\| = O\!\left(\frac{\log\beta}{\beta^2}\right), \tag{8}$$

one full order better than the naive $O(\beta^{-1})$: in the bulk $u = O(\beta|\mathcal{A}|)$, so the coupling is $O(\beta^{-2})$ there and $O(\beta^{-1})$ only in the $O(\beta^{-1})$-wide corner windows.

Proof of Lemma 6. Step 1 ((H4) makes the field linear). With the activation pattern constant, the B-field is $-G_B(\theta_B - \theta_B^*)$ with G_B the Gram matrix of the active data, hence C^∞ with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, equal to 2.13 in the reference instance. Moreover $R^{\rm hyb} = y_j - (x_j, 1)\cdot\theta_B$ is affine in θ_B and C^1 in t, with $\|\dot R^{\rm hyb}\|_\infty \le \|(x_j,1)\|\,\|G_B\|\,\mathrm{diam} =: C_R$ explicit.

Step 2 (coupling bound). The perturbation caused by A in res_i is $g_\beta(z_{A,i})$. Under (H3) the terms $i \ne j$ contribute $C_3 e^{-\beta\delta_1}$ and the term $i = j$ contributes $g_\beta(z)$; the Gram factors give a constant C_f, so $\|G_A(t)\| \le C_f\, g_\beta(z(t)) + C_3 e^{-\beta\delta_1}$.

Step 3 (integral Grönwall). Writing $\Delta = \theta_B^{\beta} - \theta_B^{\rm hyb}$ we have $\dot\Delta \le L_B\Delta + \|G_A\|$, hence $\Delta(t) \le \int e^{L_B(t-s)}\|G_A(s)\|\,ds$, and therefore

$$\sup_{[t_{\rm cross}, t_{\rm release}]} \|\Delta\| \;\le\; e^{L_B H}\left[\frac{C_f}{c\beta}\int c\log(1+w)\,dt + H\,C_3 e^{-\beta\delta_1}\right].$$

Since $g_\beta = \beta^{-1}\log(1+w)$ and Lemma 5 gives $\int c\log(1+w)\,dt = O(\log\beta/\beta)$, the total is $O(\log\beta/\beta^2)$, which is (8). Consequently $R^\beta = R^{\mathrm{hyb}} + O(\log\beta/\beta^2)$, which is the perturbation size consumed by Lemma 9. □

Verification. `prova_L2_verificacao.py` confirms that the field and R are affine with $L_B = 2.133$, that the margins of (H4) hold with $\min z_B = 0.39$, and that the refined rate holds: the metric $\sup\cdot\beta^2/\log\beta$ is stable in $[4.2, 5.7]$ over $\beta \in [3\cdot 10^2, 3\cdot 10^3]$. The exponent is what the metric discriminates: under a pure $O(\beta^{-1})$ the same metric would grow like $\beta/\log\beta$, by a factor of about 6.5 per decade, and it grew by a factor of 1.24.

A.3 The entry corner

Lemma 7 (entry corner). *Let $v = c\,|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$ be the arrival speed, transversal by (H1). The exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$ collapses the crossing-time difference to an elementary integral and gives*

$$t^\beta_{\mathrm{cross}} = t_{\mathrm{hit}} + \frac{1}{\beta v}\big(1 + O(\delta_z) + o(1)\big), \qquad \big\|\theta^\beta(t_{\mathrm{cross}}) - \theta^\infty(t_{\mathrm{hit}})\big\| = O(\beta^{-1}), \tag{9}$$

where t^β_{cross} is the first time with $w = w_0$. The exponentially weighted approach integral is $\frac{3/2}{\beta v}(1+o(1))$, the factor $3/2$ coming from $\int_0^\infty e^{-s}(1+e^{-s})\,ds$.

Proof of Lemma 7. Step 1 (pre-corner with exponential weight). For $z \ge \delta_z$ one has $|g_\beta(z) - z| \le \beta^{-1}e^{-\beta z}$ and $1-\sigma(\beta z) \le e^{-\beta z}$, so the inhomogeneity of the Grönwall comparison against the ReLU flow is $h(t) \le Ce^{-\beta z^0(t)}$. Near the hit, transversality gives $z^0(t) = v\,(t_{\mathrm{hit}} - t)(1 + O(\delta_z))$, hence

$$\int h\,dt \;\le\; C\int e^{-\beta v(t_{\mathrm{hit}}-t)}\,dt \;\le\; \frac{C}{\beta v}.$$

Along the real trajectory the weight carries in addition the factor σ from the layer, and the exact value is $\int e^{-\beta z}dt = \frac{1}{\beta v}\int_0^\infty e^{-s}(1+e^{-s})\,ds\,(1+o(1)) = \frac{3/2}{\beta v}(1+o(1))$.

The weight is self-consistent by a monotone route, with no exponential fixed point. Since $e^{-\beta z}$ is decreasing in z, the flow whose inhomogeneity is evaluated at any sub-solution $z_- \le z^\beta$ majorizes the true inhomogeneity. Iterate: $z^{(0)} = z^0$, and let $z^{(k+1)}$ solve the problem with weight $e^{-\beta z^{(k)}}$. The iteration is monotone decreasing and bounded below by the ReLU sub-solution with slack $2C/(\beta v)$: the increment of each step is at most $C/(\beta v)\cdot\sup e^{\beta(z^{(k)}-z^{(k+1)})}$, and while $|z^{(k)} - z^{(k+1)}| \le 1/\beta$ that factor is at most e, a bound maintained by the size of the increment itself, by induction on k. The iteration therefore converges to z^β with $\Delta\theta(t_\delta) = O(1/(\beta v))$, without requiring a fixed-point inequality of the form $C_2e^{c'K} \le K$.

Step 2 (window-crossing identity). On $[0, \delta_z]$ the crossing speeds are $v_0(z) = v(1 + O(\delta_z))$ for ReLU and $v_\beta(z) = v_0(z)\sigma(\beta z) + O(\beta^{-1})$ for softplus, so the difference of crossing times is

$$\int_0^{\delta_z} \Big(\frac{1}{v_\beta} - \frac{1}{v_0}\Big) dz = \frac{1}{v} \int_0^{\delta_z} \frac{1 - \sigma(\beta z)}{\sigma(\beta z)}\, dz\, (1 + O(\delta_z))$$
$$= \frac{1}{v} \int_0^{\delta_z} e^{-\beta z}\, dz\, (1 + O(\delta_z)) = \frac{1 - e^{-\beta\delta_z}}{\beta v}(1 + O(\delta_z)),$$

where the middle equality is the exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$. With $w_0 = 1$ there is no further shift, since $\log w_0 = 0$. Adding Steps 1 and 2 gives $t_{\text{cross}} - t_{\text{hit}} = \frac{1}{\beta v}(1 + O(\delta_z) + o(1))$, and the speeds are bounded on the window, so the state at crossing is within $O(\beta^{-1})$ of $\theta^\infty(t_{\text{hit}})$. □

Verification. `prova_L3_verificacao.py` confirms the two identities symbolically and both constants numerically: $K_{\text{teo}} = 1/v = 7.042$ against $K_{\text{emp}} = 7.127$ refitted from the measured crossing times, a deviation of 1.2%; and the exponentially weighted approach integral with the exact factor $3/2$ gives 10.563 in theory against 10.52 measured, below 1%.

A.4 Sticking and release for prescribed forcing

Lemma 8 (sticking and release for prescribed R). *Let $R \in C^1$ be prescribed and satisfy (H1)–(H2), and write $\mathcal{A}(t) = \int_{t_{\text{hit}}}^t R$. Then*

(i) $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + \text{corr}(t)$ with $0 \le \text{corr}(t) = \int c\log(1+w)\, ds = O(\log\beta/\beta)$; on compact subintervals of $(t_{\text{hit}}, t_{\text{exit}})$, $u = c\beta|\mathcal{A}(t)|\big(1 + O(\log\beta/(\beta a_0))\big)$, so the sticking depth is $z \sim -\beta^{-1}\log\beta$;

(ii) the release obeys the closed formula

$$t_{\text{release}} - t_{\text{exit}} = \frac{\text{corr}(\beta)}{c\,\beta\,R(t_{\text{exit}})}\,(1 + o(1)) = O\left(\frac{\log\beta}{\beta^2}\right), \tag{10}$$

so u returns to $O(1)$ at $t_{\text{exit}} + O(\beta^{-1})$, with no $\log\beta$ factor;

(iii) a perturbation $\tilde{R} = R + \delta$ shifts t_{exit} by at most $\delta\,(t_{\text{exit}} - t_{\text{hit}})/R(t_{\text{exit}}) + O(\delta^2)$.

No division by R occurs anywhere: the layer is held by the area, not by the sign of R, and the interior zero of R forced by (H1)–(H2) is harmless.

Proof of Lemma 8. Step 1 (monotone comparison). The right-hand side of $\dot\Psi = -c\beta R + c\log(1 + w(\Psi))$ is decreasing in Ψ, because $w = 1/u$ and u is increasing in Ψ. The Chaplygin comparison principle therefore applies: $R_1 \le R_2$ implies $\Psi_1 \ge \Psi_2$, and piecewise-constant envelopes of R give sub- and super-solutions that are exactly integrable through the same primitive Ψ, hence with explicit constants.

Step 2 (depth, item (i)). Integrate the exact identity (7); the correction is bounded by Lemma 5, in the same three windows. On compact subintervals of $(t_{\text{hit}}, t_{\text{exit}})$ one has

u large, so $\Psi(u) = u(1 + o(1))$ and the depth law $u = c\beta|\mathcal{A}(t)|(1 + o(1))$ follows, that is $z \sim -\beta^{-1} \log \beta$.

Step 3 (release, item (ii)). At release Ψ returns to its entry value $\Psi(u_0)$, so the constant $\Psi(u_0)$ cancels and no spurious term survives; item (i) then gives $c\beta\mathcal{A}(t_{\text{release}}) = \text{corr}(t_{\text{release}})$. By (H2) the zero of $\mathcal{A}$ at t_{exit} is transversal, with $\mathcal{A}(t) = R(t_{\text{exit}})(t - t_{\text{exit}})(1 + o(1))$, and solving for t_{release} gives (10). The delay of the release is exactly the correction integral of Lemma 5, reparametrized by the exit speed.

Step 4 (perturbation, item (iii)). Apply the implicit function theorem to $\mathcal{A}_{\tilde{R}}(t) = 0$ with denominator $R(t_{\text{exit}}) \neq 0$, splitting the episode into the same three windows: near the hit $|\mathcal{A}| \geq |R(t_{\text{hit}})|s/2$ for $s = t - t_{\text{hit}}$ small, in the bulk $|\mathcal{A}| \geq a_0(\delta)$, and near the exit the zero is transversal. This gives the shift $\leq \delta\,(t_{\text{exit}} - t_{\text{hit}})/R(t_{\text{exit}}) + O(\delta^2)$. □

Remark 13 (the corollary that explains the coupled constant). In the coupled system the initial datum comes from the entry corner, so by Lemma 7 the episode starts with a delay K/β that cuts an initial negative area $\approx R(t_{\text{hit}})K/\beta$. Item (ii) then gives $\beta\,(t_{\text{release}} - t^{\text{hyb}}_{\text{exit}}) \to -K\,|R(t_{\text{hit}})|/R(t_{\text{exit}})$. With the theoretical $K = 1/(c|R(t_{\text{hit}})|)$ this is exactly $-1/(cR(t_{\text{exit}})) = -14.81$, the coefficient of (5) left by the cancellation; with the empirical $K = 7.13$ it is $-(7.13 \times 0.2272)/0.1080 = -15.0$, against -14.38 measured, a deviation of 4% whose residue is the positive correction of item (ii).

Verification. `prova_L4_verificacao.py` measures t_{release} and corr on the same run and tests $\Delta t = \text{corr}/(c\beta R_{\text{exit}})$: the formula closes to 0.2%/1.0%/9.5% at $\beta = 3 \cdot 10^2/10^3/3 \cdot 10^3$ on a grid of $2 \cdot 10^{-6}$, the last figure having collapsed from 35% to 9.5% under a fivefold grid refinement, which identifies the residue as discretization. The corollary of Remark 13 closes at -14.99 against -14.38 measured (4.2%).

A.5 The coupled bootstrap

Lemma 9 (coupled bootstrap). *Under (H1)–(H4) the coupled system satisfies the two-term law*

$$t^{\beta}_{\text{release}} = t_{\text{exit}} - \frac{1}{c\,R^{\text{hyb}}(t_{\text{exit}})} \cdot \frac{1}{\beta} + \frac{\text{corr}(\beta)}{c\,\beta\,R^{\text{hyb}}(t_{\text{exit}})} + o(\beta^{-1}), \qquad \text{corr}(\beta) = O\left(\frac{\log \beta}{\beta}\right), \tag{11}$$

obtained by continuation with strict margins, with the constant $M_2 = e^{L_B H} C_f C_4/c$ independent of the continuation hypothesis for $\beta \geq \beta_0$. In the first-order coefficient the arrival speed cancels: in the product $K\,|R^{\text{hyb}}(t_{\text{hit}})|/R^{\text{hyb}}(t_{\text{exit}})$ of the entry delay by the area it cuts, the factor $|R^{\text{hyb}}(t_{\text{hit}})|$ cancels against $K = 1/(c|R^{\text{hyb}}(t_{\text{hit}})|)$ of Lemma 7, leaving a coefficient that still depends on c and on $R^{\text{hyb}}(t_{\text{exit}})$.

Proof of Lemma 9. The argument is a continuation with strict margins.

Step 1 (the bootstrap property). For s in $[t_{\text{cross}}, t_{\text{exit}} - \varepsilon_\beta]$ let $\mathcal{P}_s$ be the property that, on $[t_{\text{cross}}, s]$, (a) $\sup \|\theta^{\beta}_B - \theta^{\text{hyb}}_B\| < 2M_2 \log \beta/\beta^2$ and (b) $z(t) < \beta^{-1} \log w_0 - \beta^{-1}$ for $t \geq t_{\text{cross}} + C_{\text{collar}}/\beta$. The collar is necessary: at t_{cross} the margin crosses $\beta^{-1} \log w_0$

continuously, so the layer can only be strict after an $O(\beta^{-1})$ entry window. The constant is $M_2 = e^{L_B H} C_f C_4/c$, explicit through the chain Lemma 6 composed with Lemma 5.

Step 2 (nonempty and open). Lemma 7 supplies the initial datum with slack: the state is within $O(\beta^{-1})$ in θ and the layer is reached transversally, so $\mathcal{P}_s$ holds for s just past the collar. All quantities in $\mathcal{P}_s$ are continuous in s, since the trajectories are C^1, so the strict inequalities propagate to a neighbourhood.

Step 3 (improvement, hence closed). Assume $\mathcal{P}_s$ with the constant $2M_2$. Since R is affine in θ_B (Lemma 6(a)), the forcing satisfies $R^\beta = R^{\mathrm{hyb}} + \delta$ with $\|\delta\| \le \mathrm{Lip}_R \cdot 2M_2 \log\beta/\beta^2$. The sandwich of Lemma 8 with perturbed forcing then keeps $u(t) \ge c\beta(|\mathcal{A}(t)| - \|\delta\|H)/2 > u_0$ for $t \le t_{\mathrm{exit}} - \varepsilon_\beta$ with $\varepsilon_\beta = C_\varepsilon \log\beta/\beta$ and $C_\varepsilon = 4C_4/(cR_{\mathrm{exit}})$, so the layer is maintained with a margin. With the layer in hand, the Grönwall of Lemma 6 is unconditional and returns M_2, half of what was assumed, for an explicit $\beta \ge \beta_0$. The threshold β_0 is finite and traceable through the chain of constants, though we do not exhibit it. Crucially M_2 does not depend on the hypothesis $2M_2$, because the chain of constants is unconditional; this is what closes the bootstrap.

Step 4 (continuation). The set of s for which $\mathcal{P}_s$ holds is nonempty (Step 2), open (strict margins and continuity) and closed (the bounds pass to limits) in $[t_{\mathrm{cross}}, t_{\mathrm{exit}} - \varepsilon_\beta]$, hence it is the whole interval.

Step 5 (terminal window). On $[t_{\mathrm{exit}} - \varepsilon_\beta, t_{\mathrm{release}}]$ apply Lemma 8(ii)–(iii) with the forcing frozen at $R^{\mathrm{hyb}} + O(\log\beta/\beta^2)$. Three contributions add: the displacement caused by δ is $O(\log\beta/\beta^2)$ by item (iii); the correction delay is $+\mathrm{corr}/(c\beta R_{\mathrm{exit}})$ by item (ii); and the area cut at entry, $\int_{t_{\mathrm{hit}}}^{t_{\mathrm{hit}}+K/\beta} R \approx R(t_{\mathrm{hit}})K/\beta$, advances the exit by $-K|R(t_{\mathrm{hit}})|/(R(t_{\mathrm{exit}})\beta) = -1/(cR(t_{\mathrm{exit}})\beta)$, the arrival speed cancelling against $K = 1/(c|R(t_{\mathrm{hit}})|)$. Summing gives (11). □

Verification. `prova_L5_verificacao.py` checks that the two-term law reproduces the whole coupled sweep, with deviations 1.9%/0.7%/0.2%/0.1% for $\beta = 3\cdot 10^2 \ldots 10^4$, and that the surviving coefficient is $1/(cR_{\mathrm{exit}}) = 14.815$; the chain of constants is exhibited with $M_2 = 440.8$ from the empirical $C_4 = 10.6$, against $M_2 \le 1233$ from the leading theoretical term $2/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = 29.6$ alone and $M_2 \lesssim 3300$ from the full theoretical $C_4 = 80.7$, which adds the bulk term T_{ep}/a_0 with $T_{\mathrm{ep}} = t_{\mathrm{exit}} - t_{\mathrm{hit}} = 1.021$ and $a_0 \approx 0.02$ in the instance. The strict layer is confirmed at $\beta = 10^4$. One honest caveat: with the conservative C_ε the inner window of the layer check is empty for $\beta \le 3\cdot 10^3$, so the check is vacuous there, and at those β the maintenance of the layer is implied only by the detection of the release itself.

A.6 The exit corner, branch selection and the scalar reduction

Lemma 10 (exit corner, branch selection and scalar reduction). *Under (H1)–(H4), after release: (i) the β-trajectory crosses the exit corner in time $\frac{1}{\beta v'}(1 + o(1))$ with $v' = c\,R^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, the mirror of Lemma 7; (ii) the ReLU flow restarted at $(\theta_A(t_{\mathrm{hit}}), \theta_B^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ has forward uniqueness on the activating branch, and that branch is selected by the limit itself, the frozen Filippov branch [7] being incompatible with release at $t_{\mathrm{exit}} + O(\beta^{-1})$; (iii) classical*

Grönwall glues on $[t_{\mathrm{release}}, H]$ *at rate* $O(\log\beta/\beta)$. *Under* (H3) *the other data contribute* $O(e^{-\beta\delta_1})$, *the face-j field is rank one along* $(x_j, 1)$, *and the episode reduces to the scalar problem.*

Proof of Lemma 10. Step 1 (item (i)). Mirror Lemma 7, replacing v by $v' = cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, which is transversal by (H2). The crossing identity is the same, and the weight is $\frac{3/2}{\beta v'}$.

Step 2 (item (ii), forward uniqueness and branch selection). At the restart point the active field is locally Lipschitz on the side $z \geq 0$ and $\dot{z}(t_{\mathrm{exit}}^+) = v' > 0$, so the solution enters $\{z > \epsilon t\}$ and Picard uniqueness holds forward from any $t > t_{\mathrm{exit}}$. The only Filippov alternative [7], namely remaining in $z \leq 0$ where the field is identically zero, is incompatible with the limit of the θ^β, since Lemma 9 places the release at $t_{\mathrm{exit}} + O(\beta^{-1})$. The selection of the activating branch is therefore proved by Lemma 9 and not postulated in Definition 1.

Step 3 (item (iii), gluing). The initial datum of the gluing satisfies $\|\theta^\beta(t_{\mathrm{release}}) - \theta^\infty(t_{\mathrm{exit}})\| = O(\beta^{-1}\log\beta)$: the position of θ_A in the layer contributes $O(\log\beta/\beta)$, θ_B contributes the rate of Lemma 6, and the time displacement $O(\beta^{-1})$ contributes through bounded speeds. In the active interior the two fields are exponentially close, so classical Grönwall on $[t_{\mathrm{release}}, H]$ keeps the gap at $O(\log\beta/\beta)$.

Step 4 (scalar reduction). The field on θ_A is $-\frac{1}{n}\sum_i \operatorname{res}_i \sigma(\beta z_i)(x_i, 1)$. Under (H3), that is $z_{A,i} \leq -\delta_i < 0$ for $i \neq j$ on the limit episode, the terms $i \neq j$ are bounded by $Ce^{-\beta\delta_i}$ and are absorbed by the Grönwall estimates without changing any $O(\beta^{-1})$ rate. The term $i = j$ is rank one along $(x_j, 1)$, so the component of θ_A tangential to the face is frozen at the exponential rate, while the normal component is exactly the scalar problem of Lemmas 5 to 9. Hypothesis (H3) is an open condition on the data and the initialization, and it is verified numerically in the instance with an explicit margin δ_1. □

Verification. `prova_L6_verificacao.py` measures the gluing directly: for $\beta \in \{10^3, 3\cdot 10^3\}$ the supremum of $\|\theta^\beta - \theta^{\mathrm{relu}}\|$ over $[t_{\mathrm{release}} + 0.05, H]$ is $1.6\cdot 10^{-3}$ and $5.5\cdot 10^{-4}$, so the metric $\sup\cdot\beta/\log\beta$ is stable at 0.21–0.23, confirming the rate $O(\log\beta/\beta)$. `passo6_verificacao.py` confirms the scalar reduction: (H3) holds with $\delta_1 = 0.508$ and the tangential drift is below 10^{-10}, machine zero, at both values of β, consistent with $e^{-\beta\delta_1} \sim 10^{-65}$.

The debugging of this script is worth recording, because it exhibits the alternative branch. The first version of the reference restarted the ReLU flow at $z_A = -1.1\cdot 10^{-6}$, an overshoot of one Euler step of the pre-hit loop, and fell into the frozen Filippov branch, with a constant gap of ≈ 0.039. The correct reference requires exact projection onto the face together with an activating nudge larger than the overshoot. Branch selection is a real condition of the statement.

B Algorithms

Four routines carry the experimental claims of Section 6: the ground-truth area clock (Algorithm 1), the truncated calibrated null (Algorithm 2), the hysteresis counter (Algorithm 3) and the pre-registered decision procedure (Algorithms 4 and 5). They are given here in full,

because each of them is an instrument built for this paper and each of them has already falsified one of its own predecessors. Smaller routines are folded into the algorithm that consumes them: the clean-episode filter inside Algorithm 2, the Clopper–Pearson bisection and the gate n^* inside Algorithms 4 and 5.

B.1 The ground-truth area clock

The ruler of Section 6.2 was replaced, after it failed its own null, by a clock with no ruler at all: death and return are read directly off the margin over all data, with no threshold, no debounce and no dwell. The single free choice left is the accumulator, and it runs on the same clock, so the temporal ordering $t_{\text{hit}} \le t_{\text{area}} < t_{\text{res}}$ is structural rather than measured; the only empirical failure mode left is a return with no area.

One normalization is deliberate and is recorded here because it looks like an inconsistency. The descent step is taken on $L = \frac{1}{2n}\sum_j \text{res}_j^2$, so the parameter move carries a factor $1/n$, whereas the accumulated drive $\hat{\mathcal{A}}_k$ is built from the leader-datum quantity $\hat{R}_k$, which does not. The two differ by the fixed positive constant $1/n$, and every claim read off the clock — the sign of $\hat{\mathcal{A}}_k$, the first instant at which it turns positive (t_{area}) and hence the ordering $t_{\text{hit}} \le t_{\text{area}} < t_{\text{res}}$ — is invariant under multiplication by a positive constant. What is *not* invariant is the magnitude of $\hat{\mathcal{A}}_k$, which is therefore never compared against the theoretical area of Theorem 2 and is used only through its sign and its first zero crossing.

Two guards are asserted before any run is used. Bit-exact parity: the loss trace must agree with the pilot engine to 10^{-12}, so the clock is known to be a pure observer. And the episode count must equal the number of born-dead units plus the number of dynamic deaths, so no episode is silently dropped.

Executable: `pilotos-4a2/pressao/taxa/motor_4a3.py`, *config SHA in* `results-4a3/manifest-4a3.json`.

B.2 The truncated calibrated null

This is the methodological repair of Section 6.2, and it is the one piece of the paper we would offer for reuse. The instrument stops measuring the area at the resurrection, so $t_{\text{area}} \le t_{\text{res}}$ holds mechanically and a naive permuted-delay placebo, which does not inherit that truncation, inflates the false-positive rate by up to $58\times$ the nominal α. The repair is to put the truncation *inside* the null: fit the delay distribution treating non-firing as right censoring, then reproduce the truncation in every bootstrap replicate, so the structural bias sits on both sides of the comparison and cancels.

Under H_0 the delay is independent of the duration, which is exactly what the resampling imposes: the durations are never resampled, only the delays are, and they are drawn from the pooled $\hat{F}$. The level and the power are demonstrated on synthetic truncated worlds before the procedure is applied to data.

Executable: `etapa4-mlp-populacao/analise.py` *(*`p_calibrado`*,* `_km`*,* `_amostra_km`*),* *config SHA in* `etapa4-mlp-populacao/results/manifest.json`.

Algorithm 1 Ground-truth area clock (deterministic MLP, $T = 0$).

Require: width m, activation g_β, rate η, horizon H, seed; data (x, y) with n points
Ensure: for every episode: $(k, \text{class}, t_{\text{death}}, t_{\text{area}}, t_{\text{res}}, \text{revived}, z_{\text{park}})$

initialize (w, b, v) from the seed; $\text{open}_k \leftarrow$ FALSE for all k
for $t = 0$ **to** $H - 1$ **do**
  $z \leftarrow xw^\top + b$; $a \leftarrow g_\beta(z)$; $f \leftarrow av$; $\text{res} \leftarrow f - y$
  $z_k^{\max} \leftarrow \max_{j \le n} z_{jk}$; $j_k^* \leftarrow \arg\max_{j \le n} z_{jk}$
  $\text{dead}_k \leftarrow [\, z_k^{\max} < 0 \,]$ {ground truth: the unit is off on *every* datum}
  $\hat{R}_k \leftarrow v_k \big[(y - f)_{j_k^*} + v_k \, a_{j_k^* k} \big]$ {leader-datum residual drive; the $1/n$ of L is dropped (see below)}
  for all k with $\text{dead}_k \wedge \neg\text{open}_k$ **do**
    open an episode: $\text{open}_k \leftarrow$ TRUE, $\hat{\mathcal{A}}_k \leftarrow 0$, $t^k_{\text{area}} \leftarrow -1$, $t^k_{\text{death}} \leftarrow t$, $z^k_{\text{park}} \leftarrow z_k^{\max}$
    class $\leftarrow$ BORN if $t = 0$, else DYN
  end for
  for all k with $\text{open}_k \wedge \text{dead}_k$ **do**
    $\hat{\mathcal{A}}_k \leftarrow \hat{\mathcal{A}}_k + \hat{R}_k \, \eta$ {the area accrues only on dead steps}
    if $t^k_{\text{area}} < 0$ **and** $\hat{\mathcal{A}}_k > 0$ **then**
      $t^k_{\text{area}} \leftarrow t$ {the gate opens}
    end if
    $z^k_{\text{park}} \leftarrow \min(z^k_{\text{park}}, z_k^{\max})$
  end for
  for all k with $\text{open}_k \wedge (z_k^{\max} > 0)$ **do**
    close the episode with $t_{\text{res}} \leftarrow t$, revived $\leftarrow$ TRUE; $\text{open}_k \leftarrow$ FALSE
  end for
  $\theta \leftarrow \theta - \eta \nabla_\theta \frac{1}{2n} \sum_j \text{res}_j^2$ {plain gradient descent on the loss L of Section 2; the instrument never touches the dynamics}
end for
close every still-open episode as **censored**: $t_{\text{res}} \leftarrow$ NONE, revived $\leftarrow$ FALSE {never recorded as "permanent"}

B.3 The hysteresis counter

In the transformer the input of the block is non-stationary, so the margin chatters across zero and a bare threshold counts noise as resurrection. The counter therefore commits a transition only outside a band $\pm\delta_h$ calibrated from data, and a return counts only if the unit is still alive at the horizon.

Inside the band $[-\delta_h, +\delta_h]$ the state is held, which is what kills the chatter; the test that the band is doing work is that the return count at $\delta_h = 0.05$ is strictly below the count at $\delta_h = 0$ on a seed known to chatter. One disclosure is on record: under TRAIN the same calibration gives $\delta_h \approx 0.07$–0.08, about $1.7\times$ larger, so every estimand that depends on δ_h being decidable is confined to the FROZEN cells.

Executable: `etapa5-transformer/motor_e5_conf.py`, *config SHA in* `results-e5/`

Algorithm 2 Truncated calibrated null for the area-delay statistic.

Require: episodes grouped by seed; B replicates; cell key (i_b, i_m)
Ensure: p, T_{obs}, effective number of seeds

Retention (clean-episode filter). Keep an episode iff it completed ($t_{\text{res}} \geq 0$) and: at most q switching events of *other* units in $(t_{\text{death}}, t_{\text{res}}]$; the readout v_k did not change sign; $\min_t |v_k| \geq \varphi_v$, which excludes death by a vanishing readout; and no change of the leader j^*. Drop seeds with fewer than three surviving episodes.
for each episode: duration $d_i \leftarrow t_{\text{res}} - t_{\text{death}}$; delay $a_i \leftarrow t_{\text{area}} - t_{\text{death}}$ if the area fired, else -1; $f_i \leftarrow [\, a_i \geq 0 \,]$
Survival encoding. $\text{obs}_i \leftarrow a_i$ if f_i else d_i; $\text{ev}_i \leftarrow f_i$ {non-firing = right censoring at the episode's own duration}
$T_{\text{obs}} \leftarrow \textsc{Stat}(\{(d, a, f)\})$ with one independent stream per seed
$\hat{F} \leftarrow \textsc{KaplanMeier}(\text{obs}, \text{ev})$, pooled over the seeds of the cell
exceed $\leftarrow 0$
for $b = 1$ **to** B **do**
  for all seeds, holding the durations d *fixed* **do**
    draw $A_i \leftarrow \hat{F}^{-1}(u_i)$, $u_i \sim U[0,1)$, and set $A_i \leftarrow \infty$ when u_i exceeds $\hat{F}$ at the last observed event {the censored mass stays censored}
    $f_i^\dagger \leftarrow [\, A_i \leq d_i \,]$; $a_i^\dagger \leftarrow A_i$ if $f_i^\dagger$ else -1 {the truncation is reproduced, replicate by replicate}
  end for
  $T_b \leftarrow \textsc{Stat}(\{(d, a^\dagger, f^\dagger)\})$; exceed $+= [\, T_b \geq T_{\text{obs}} \,]$
end for
return $p = (1 + \text{exceed})/(B + 1)$

function STAT: for each seed, restrict to the fired episodes; if fewer than two, the seed contributes nothing. Otherwise draw for each i an index j_i uniform on the other fired episodes of that seed and put $W = \frac{1}{n} \sum_i \big[\, (d_i - a_i) < |d_i - a_{j_i}| \,\big]$. Return the mean of W over the surviving seeds.

manifest-e5.json.

B.4 The pre-registered decision procedure

The same procedure serves Section 6.4 and Section 6.5, with the estimand-specific thresholds read from the frozen config. Its one structural commitment is the last line: insufficient evidence returns INCONCLUSIVE and never a negative.

The thresholds are derived, not chosen: ϑ is half the pilot's Clopper–Pearson lower bound on the same proportion, and τ is the pilot's $q_{0.9}$ return time. Both are read from the config committed before the grid ran. One honest disclosure: at $N = 50$ the transformer's KILLS branch is nearly unreachable, since the Clopper–Pearson upper bound on $0/50$ already exceeds ϑ; the operative decision there is TRANSFERS against INCONCLUSIVE, and it would

take $N \gtrsim 800$ for the kill to become available.
Executables: `pilotos-4a2/pressao/taxa/analise_4a3.py` *and* `etapa5-transformer/analise_e5.py`, *config SHAs in* `results-4a3/manifest-4a3.json` *and* `results-e5/manifest-e5.json`.

C Pre-registered estimands of the transformer confirmatory

The following five estimands were frozen at commit `1b45498`, ahead of the data, and are reproduced here verbatim; Section 6.6 carries the two scoping findings S1 and S2 that motivated them.

- **E-A, validity.** Any sustained return in the relu-frozen arm will invalidate the run, since an own-gradient identically zero together with a fixed input makes death terminal. The pilot gave 0/40. *Result:* **VALID.** 0/100 sustained returns in relu-frozen across both injection depths.

- **E-B, upstream channel.** We will estimate $\Delta = p_{\text{sust}}(\text{train}) - p_{\text{sust}}(\text{frozen})$ per arm, and will call the effect reproduced if the bootstrap lower bound is positive. Under the hysteresis instrument the test is stricter than in the pilot. The hysteresis band was calibrated on frozen runs only; alive units in training runs fluctuate up to 1.7× wider, so in the training arms the band is narrow relative to the arm's own noise and crossings are easier — an anti-conservative direction for Δ. The sustained-return filter is the measured containment: five returns failed sustainment (hover), all five in training cells and none in frozen, and none entered the counts. The ReLU arm is, in addition, immune to that argument: its own-move is identically zero on both sides of the contrast, so if a band that is narrow relative to training noise were manufacturing spurious crossings, they would appear in relu-frozen as well — and relu-frozen gives 0/100. The 20/100 against 0/100 contrast (two-sided Fisher exact test, $p = 6.6 \cdot 10^{-7}$) is therefore the internal control that closes E-B without relying on the calibration of δ_h. *Result:* **REPRODUCED in all four arms.** $\Delta = 0.210\,[0.120, 0.300]$ (sp-β3), $0.200\,[0.110, 0.280]$ (sp-β10), $0.220\,[0.120, 0.320]$ (GELU), $0.200\,[0.120, 0.280]$ (ReLU, 20/100 against 0/100): the non-stationary FFN input is a real resurrection channel, of comparable size across activations.

- **E-C, transfer of the theorem class.** For sp-β3 frozen under shallow injection we will estimate $p(\text{sustained return} \mid \text{injection})$, and will call the class transferred if there are at least $n^* = 6$ sustained returns. This is the direct analogue of C2 in the MLP. *Result:* **TRANSFERRED.** 22/50 sustained returns, $\hat{p} = 0.440\,[0.300, 0.587]$, 3.7× the gate $n^* = 6$; the area of the leader drive opened before the return in 21/22 of them. The memory gate of the theorem operates in a transformer FFN through the own channel.

- **E-D, GELU barrier.** For gelu-frozen we will compare deep injection (-2.0) against shallow (-0.3); the barrier predicts shallow far above deep in the own channel, with a

depth-matched control against sp-β3. The pilot at -1.0 was a tie and did not decide. *Result:* **Consistent with the barrier, confounded with depth** (the three-way label frozen at pre-registration). Intra-GELU: shallow 29/50 vs. deep 0/50; but the depth-matched control sp-β3-frozen at -2.0 also gave 0/50, so depth alone accounts for the deep zero and the sustained-barrier claim is not licensed by this design. The S1 signature remains sharp: $\text{sign}(\mathcal{A}_{\text{drive}}) \neq \text{sign}(\mathcal{A}_{\text{mlp}})$ in 92% of deep vs. 4% of shallow episodes — the barrier is present in the drive field; its terminality is not separable from depth here. The floor is structural, not accidental: at $z = -2.0$ the own channel of sp-β3 is $\sigma(3z) = 0.0025$, that is 0.9% of its value at $z = -0.3$ (gradient starvation), while $\text{GELU}'(-2.0) \approx -0.085$ is adverse — the depth -2.0 is deep for *both* mechanisms at once, so 0/50 against 0/50 (two-sided Fisher exact test, $p = 1.0$) carries no information by construction. The design that would separate them injects at $z \approx -0.9$, where $\text{GELU}' \approx -0.055$ is already adverse but $\sigma(3z) \approx 0.063$ still leaves $\approx 22\%$ of the shallow softplus channel; a sweep over $\{-0.5, -0.75, -0.9, -1.2\}$ tests the differential prediction — an abrupt drop for GELU at $z^* = -0.7518$ against a smooth decay for softplus — a signature no depth confound reproduces. This is inherited as future work, not claimed here.

- **E-E, β staircase.** We will estimate p(sustained return) across $\beta \in \{3, 10\}$, frozen and training, as an estimation with no kill condition. *Result:* $\beta = 3$ above $\beta = 10$ in both factors (frozen $0.220\,[0.143, 0.314]$ against $0.090\,[0.042, 0.164]$; training $0.430\,[0.331, 0.533]$ against $0.290\,[0.204, 0.389]$). GELU itself is the most revivable arm (shallow frozen $0.580\,[0.432, 0.718]$; with the upstream channel $0.860\,[0.733, 0.942]$): the activation used in practice sits at the open end of the channel.

D Post-hoc aggregation of C3

C3 was pre-registered as estimation-only, with no kill condition, and the frozen analysis script contains no aggregate for it (Section 6.4). The paragraph below was produced after the confirmatory data were seen, at the request of an external reviewer, and is exploratory throughout: nothing in it tests a pre-registered hypothesis, and with 13 episodes over 10 seeds the rank intervals are wide by construction.

In 10 of the 13 episodes with an open area the gate is open at the moment of death, $t_{\text{area}} = t_{\text{death}}$ exactly, so in those episodes the lead time $t_{\text{res}} - t_{\text{area}}$ is identical to the duration of the episode and carries no information of its own. Controlling for when the episode began, the partial Spearman correlation between t_{area} and t_{res} given t_{death} is $+0.10$, with a 95% seed-clustered bootstrap interval $[-0.43, +0.78]$ that crosses zero: there is no evidence that the zero of the area predicts the return time beyond the onset of the episode. Under this clock the gate does not warm up; in most episodes it is already open at death, and what varies is how long the rise takes. The mean drive of the episode is not recoverable from the instrument, which persists only the triple of times, z_{park}

and the channel value, so the question cannot be pushed further with these runs. Persisting the drive series per episode is inherited as a requirement for the next stage.

Generated by `agrega_c3_posthoc.py` in the confirmatory's directory.

Algorithm 3 Hysteresis counter with sustained return (transformer block).

Require: arm, attention factor TRAIN/FROZEN, injection depth $\kappa \in \{0.3, 2.0\}$, band δ_h, horizon H, seed

Ensure: per unit: $(\text{revived}, \text{sustained}, \text{hover}, t_{\text{res}}, t_{\text{area}}, \mathcal{A}_{\text{drive}}, \mathcal{A}_{\text{mlp}}, \text{own-move}, \text{channel})$

Calibration of δ_h (offline, once). Run the frozen flow with no injection; on units alive throughout, take the half peak-to-peak amplitude of $z^{\max}$ in windows of 200 steps and set δ_h to the $q_{0.99}$ of that amplitude. This gives $\delta_h = 0.0476$, and it is frozen in the config before the grid runs.

Injection. Compute $z_k^{\max,0}$ and pick the unit k of lowest initial margin; shift $b_1[k]$ by $-\kappa - z_k^{\max,0}$, so that $z_k^{\max} = -\kappa$ exactly. The depth is a two-level factor across the grid, shallow -0.3 against deep -2.0, straddling the GELU sign-flip barrier $z^* = -0.752$.

$\text{alive}_k \leftarrow$ TRUE for all k

for $t = 0$ **to** $H - 1$ **do**

  forward pass; $z_k^{\max} \leftarrow \max_j z_{jk}$ over all $N \cdot S$ tokens; $j_k^* \leftarrow \arg\max_j z_{jk}$

  for all k with $\text{alive}_k \wedge (z_k^{\max} < -\delta_h)$ **do**

    confirmed death: $\text{alive}_k \leftarrow$ FALSE; open an episode and snapshot $(W_1[k], b_1[k])$

  end for

  for all k with an open episode **do**

    $c_k \leftarrow (w_{\text{out}} W_2)_k$; $\mathcal{A}_k^{\text{mlp}} \mathrel{+}= c_k\big[(y - \hat{y})_{j_k^*} + c_k a_{j_k^* k}\big]\eta$; $\mathcal{A}_k^{\text{drive}} \mathrel{+}= c_k (y - \hat{y})_{j_k^*} \, g'_\beta(z_k^{\max})\, \eta$

    record the first t at which each area turns positive

  end for

  for all k with $\neg\text{alive}_k \wedge (z_k^{\max} > +\delta_h)$ **do**

    confirmed return: $\text{alive}_k \leftarrow$ TRUE, $t_{\text{res}} \leftarrow t$; close the episode

    own-move $\leftarrow \|(W_1[k], b_1[k]) - (W_1, b_1)_{\text{death}}\|_2$; channel $\leftarrow$ UPSTREAM if own-move $< 10^{-9}$, else OWN

  end for

  gradient step on the trainable parameters only (under FROZEN, attention and the readout are held fixed)

end for

close every open episode as censored, and per unit set sustained $\leftarrow \text{revived} \wedge \text{alive}_k$, hover $\leftarrow \text{revived} \wedge \neg\text{alive}_k$

Algorithm 4 Pre-registered decision procedure (part 1 of 2): classification, eligibility and gate.

Require: runs; horizon H; return-time quantile $\tau = q_{0.9}$ of the pilot; gate n^*; kill threshold ϑ; floor π_0; $\alpha = 0.05$

Ensure: per estimand, one of REPRODUCES/SUSTAINS/KILLS/INCONCLUSIVE (for the barrier estimand the three-way label SUSTAINS/CONSISTENT-CONFOUNDED/KILLS of the depth-matched rule)

1: **Classification.** For each dynamic death:

- *return* if the unit revived on the ground-truth clock; it is *ordered* if moreover $t_{\text{death}} \leq t_{\text{area}} < t_{\text{res}}$ with $t_{\text{area}} \geq 0$;
- *censored* if it did not revive but its area had opened;
- *terminal* if it did not revive and its area never opened.

2: **Eligibility.** The death is eligible iff $t_{\text{death}} \in [0, H - \tau]$, the only window in which a typical return still fits inside the horizon. In the transformer the injected deaths all occur at $t = 0$, so there the window is applied to t_{res} instead and yields a secondary robustness count, not the primary estimand.

3: **Gate.** $n^* \leftarrow$ the α-quantile of Poisson(μ), with μ the expected number of deaths implied by the pilot's bootstrap lower bound on the death rate; here $n^* = 10$ for the MLP and 6 for the transformer. It is fixed before the run.

Algorithm 5 Pre-registered decision procedure (part 2 of 2): intervals, estimands and the validity control.

Interval. $(\hat{p}, \ell, u) \leftarrow$ ClopperPearson(k, N, α), where ℓ solves $\Pr[X \geq k \mid \mathrm{Bin}(N, p)] = \alpha/2$ and u solves $\Pr[X \leq k \mid \mathrm{Bin}(N, p)] = \alpha/2$, each by 200 bisection steps on the exact binomial tail; $\ell = 0$ when $k = 0$ and $u = 1$ when $k = N$.
Rate estimand (k eligible returns out of N eligible deaths):
if $N < n^*$ **then**
  return INCONCLUSIVE {never negative on insufficient N}
else if $u < \vartheta$ **then**
  return KILLS
else
  return REPRODUCES
end if
Ordering estimand (k ordered out of N returns):
if $N < n^*$ **then**
  return INCONCLUSIVE
else if $\ell < \pi_0$ **then**
  return KILLS
else
  return SUSTAINS
end if
Validity control. If any run of the zero-gradient arm (ReLU in the MLP, relu-frozen in the transformer) records a return, declare the whole run **invalid**; no verdict on any substantive estimand may then be claimed.
Staircase estimands are reported with intervals and carry no kill condition, by pre-registration.